

Immunohematology

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4.1

Genetics and Immunology of Blood Groups

1. **What type of serological testing does the blood bank technologist perform when determining the blood group of a patient?**

A. Genotyping
B. Phenotyping
C. Both genotyping and phenotyping
D. Polymerase chain reaction

Blood bank/Apply knowledge of laboratory operations/Genetics/1

2. **If anti-K reacts 3+ with a donor cell with a genotype KK and 2+ with a Kk cell, the antibody is demonstrating:**

A. Dosage
B. Linkage disequilibrium
C. Homozygosity
D. Heterozygosity

Blood bank/Apply knowledge of fundamental biological characteristics/Genetics/Kell/3

3. **Carla expresses the blood group antigens Fy^a, Fy^b, and Xg^a. James shows expressions of none of these antigens. What factor(s) may account for the absence of these antigens in James?**

A. Gender
B. Race
C. Gender and race
D. Medication

Blood bank/Apply knowledge of fundamental biological characteristics/Genetics/2

4. **Which of the following statements is true?**

A. An individual with the BO genotype is homozygous for B antigen
B. An individual with the BB genotype is homozygous for B antigen
C. An individual with the OO genotype is heterozygous for O antigen
D. An individual with the AB phenotype is homozygous for A and B antigens

Blood bank/Apply knowledge of fundamental biological characteristics/Genetics/ABO/1

5. **Which genotype is heterozygous for C?**

A. DCe/dce
B. DCE/DCE
C. Dce/dce
D. DCE/dCe

Blood bank/Apply knowledge of fundamental biological characteristics/Genetics/Rh/2

Answers to Questions 1–5

1. **B** Phenotyping, or the physical expression of a genotype, is the type of testing routinely

performed in the blood bank. An individual, for example, may have the AO genotype but phenotypes as group A.

2. **A** Dosage is defined as an antibody reacting stronger with homozygous cells (e.g., KK) than with heterozygous cells (e.g., Kk). In addition to Kell, dosage effect is seen commonly with antigens M, N, S, s, Fy^a, Fy^b, Jk^a, Jk^b, and the antigens of the Rh system.
3. **C** The frequency of Duffy antigens Fy^a and Fy^b varies with race. The Fy(a-b-) phenotype occurs in almost 70% of African Americans and is very rare in whites. Xg^a antigen is X-linked and, therefore, expressed more frequently in women (who may inherit the antigen from either parent) than in men.
4. **B** An individual having the BB genotype has inherited the B gene from both parents and, therefore, is homozygous for B antigen.
5. **A** The genotype DCE/dce contains one C and one c gene and is heterozygous for C and c antigens.

6. Which genotype(s) will give rise to the Bombay phenotype?

- A. HH only
- B. HH and Hh
- C. Hh and hh
- D. hh only

Blood bank/Apply knowledge of fundamental biological characteristics/ABO/Bombay/1

7. Meiosis in cell division is limited to the ova and sperm producing four gametes containing what complement of DNA?

- A. 1N
- B. 2N
- C. 3N
- D. 4N

Blood bank/Apply knowledge of fundamental biological characteristics/Genetics/1

8. A cell that is not actively dividing is said to be in:

- A. Interphase
- B. Prophase
- C. Anaphase
- D. Telophase

Blood bank/Apply knowledge of fundamental biological characteristics/Genetics/1

9. Which of the following describes the expression of most blood group antigens?

- A. Dominant
- B. Recessive
- C. Codominant
- D. Corecessive

Blood bank/Apply knowledge of fundamental biological characteristics/Genetics/1

10. What blood type is not possible for an offspring of an AO and BO mating?

- A. AB
- B. A or B
- C. O

D. All are possible

Blood bank/Apply knowledge of fundamental biological characteristics/Genetics/ABO/2

11. The alleged father of a child in a disputed case of paternity is blood group AB. The mother is group O, and the child is group O. What type of exclusion is this?

A. Direct/primary/first order

B. Probability

C. Random

D. Indirect/secondary/second order

Blood bank/Evaluate laboratory data to verify test results/Genotype/Paternity testing/2

12. If the frequency of gene Y is 0.4 and the frequency of gene Z is 0.5, one would expect that they should occur together 0.2 (20%) of the time. In actuality, they are found together 32% of the time. This is an example of:

A. Crossing over

B. Linkage disequilibrium

C. Polymorphism

D. Chimerism

Blood bank/Apply principles of genetics/3 Answers to Questions 6–12

Answers to Questions 6–12

6. D The Bombay phenotype will be expressed only when no H substance is present. The O_h type is expressed by the genotype hh . Bombay phenotypes produce naturally occurring anti-H, and their serum agglutinates group O red blood cells (RBCs) in addition to RBCs from persons who are groups A, B, and AB.

7. A Meiosis involves two nuclear divisions in succession resulting in four gametocytes, each containing half the number of chromosomes found in somatic cells or $1N$.

8. A Interphase is the stage in between cell divisions. The cell is engaged in metabolic activity. Chromosomes are not clearly discerned; however, nucleoli may be visible.

9. C The inheritance of most blood group genes is codominant, meaning that no gene or allele is dominant over another. For example, a person who is group AB expresses both the A and B antigen on his or her RBCs.

10. D A mating between AO and BO persons can result in an offspring with blood type A, B, AB, or O.

11. D An indirect/secondary/second order, exclusion occurs when a genetic marker is absent in the child but should have been transmitted by the alleged father. In this case, either A or B should be present in the child.

12. B Linkage disequilibrium is a phenomenon in which alleles situated in close proximity on a chromosome associate with one another more than would be expected from individual allelic frequencies.

13. In the Hardy-Weinberg formula, p^2 represents:

A. The heterozygous population of one allele

B. The homozygous population of one allele

- C. The recessive allele
- D. The dominant allele

Blood bank/Apply knowledge of fundamental biological characteristics/Genetics/1

- 14. In this type of inheritance, the father carries the trait on his X chromosome. He has no sons with the trait because he passed his Y chromosome to his sons; however, all his daughters will express the trait.**
- A. Autosomal dominant
 - B. Autosomal recessive
 - C. X-linked dominant
 - D. X-linked recessive

Blood bank/Apply knowledge of fundamental biological characteristics/Genetics/1

- 15. Why do immunoglobulin M (IgM) antibodies, such as those formed against ABO antigens, have the ability to directly agglutinate RBCs and cause visible agglutination?**
- A. IgM antibodies are larger molecules and have the ability to bind more antigen
 - B. IgM antibodies tend to clump together more readily to bind more antigen
 - C. IgM antibodies are found in greater concentrations than IgG antibodies
 - D. IgM antibodies are not limited by subclass specificity

Blood bank/Apply knowledge of fundamental biological characteristics/Antibodies/1

- 16. Which of the following enhancement mediums decreases the zeta potential, allowing antibody and antigen to come closer together?**
- A. Low ionic strength solution (LISS)
 - B. Polyethylene glycol
 - C. Polybrene
 - D. Dithiothreitol-papain (ZZAP)

Blood bank/Apply knowledge of fundamental biological characteristics/Antigens/1

- 17. This type of antibody response is analogous to an anamnestic antibody reaction.**
- A. Primary
 - B. Secondary
 - C. Tertiary
 - D. Anaphylactic

Blood bank/Apply knowledge of fundamental biological characteristics/Antibodies/1

- 18. Which antibodies to a component of complement are contained in the rabbit polyspecific anti-human globulin (AHG) reagent for detection of in vivo sensitization?**
- A. Anti-IgG and anti-C3a
 - B. Anti-IgG and anti-C3d
 - C. Anti-IgG and anti-IgM
 - D. All of these options

Blood bank/Apply knowledge of fundamental biological characteristics/AHG/2

Answers to Questions 13–18

- 13. B** In the Hardy-Weinberg formula ($p^2 + 2pq + q^2$), p^2 and q^2 represent homozygous expressions, and $2pq$ represents heterozygous expression. This formula is used in population genetics to determine the frequency of different alleles.
- 14. C** In X-linked dominant inheritance, there is absence of male-to-male transmission

because a male passes his Y chromosome to all of his sons and his single X chromosome to all his daughters. All daughters who inherit the affected gene will express the trait. An example of this type of inheritance is the X^g_a blood group.

- 15. A** An IgM molecule has the potential to bind up to 10 antigens compared with a molecule of IgG, which can bind only two.
- 16. A** LISS contains a reduced concentration of NaCl (0.2%) and results in a reduction in charged ions within the ionic cloud, decreasing the zeta potential and facilitating antigen–antibody interaction.
- 17. B** An anamnestic response is a secondary immune response in which memory lymphocytes respond rapidly to foreign antigen in producing specific antibody. The antibodies are IgG and are produced at lower doses of antigen than in the primary response.
- 18. B** In the direct antiglobulin test (DAT), rabbit polyspecific antisera contains both an anti-human IgG component and an antibody against the C3d component of complement.

4.2

ABO Blood Group System

1. Which of the following distinguishes the A₁ blood group from the A₂ blood group?

- A. A₂ antigen will not react with anti-A, A₁ will react strongly (4+)
- B. An A₂ person may form anti-A₁; an A₁ person will not form anti-A₁
- C. An A₁ person may form anti-A₂, an A₂ person will not form anti-A₁
- D. A₂ antigen will not react with anti-A from a nonimmunized donor; A₁ will react with any anti-A

Blood bank/Apply knowledge of fundamental biological characteristics/ABO blood group/2

2. A patient's serum is incompatible with O cells. The patient RBCs give a negative reaction to anti-H lectin. What is the most likely cause of these results?

- A. The patient may be a subgroup of A
- B. The patient may have an immunodeficiency
- C. The patient may be a Bombay phenotype individual
- D. The patient may have developed alloantibodies

Blood bank/Apply principles of special procedures/ABO blood group/3

3. What antibodies are formed by a Bombay phenotype individual?

- A. Anti-A and anti-B
- B. Anti-H
- C. Anti-A,B
- D. Anti-A, B, and H

Blood bank/Apply knowledge of fundamental biological characteristics/ABO blood group/Bombay/1

4. Acquired B antigens have been found in:

- A. Bombay phenotype individuals
- B. Group O persons
- C. Persons of all blood groups
- D. Group A persons

Blood bank/Apply knowledge of fundamental characteristics/ABO/1

5. Blood is crossmatched on an A-positive person with a negative antibody screen. The patient received a transfusion of A-positive RBCs 3 years ago. The donors chosen for crossmatching were A-positive. Crossmatching was run on the automated analyzer and yielded 3+ incompatibility. How can these results be explained?

- A. The patient has an antibody to a low-frequency antigen
- B. The patient has an antibody to a high-frequency antigen
- C. The patient is an A₂ with anti-A₁
- D. The patient is an A₁ with anti-A₂

Blood bank/Apply principles of special procedures/ABO/3

Answers to Questions 1–5

1. **B** The group A₁ comprises both A₁ and A antigens. Anti-A will react with both A₁- and A₂-positive RBCs. A person who is group A₂ may form anti-A₁, but an A₁ person will not form anti-A₁ (which would cause autoagglutination).
2. **C** Bombay phenotype is the only ABO phenotype incompatible with O cells. The RBCs of a Bombay phenotype individual show a negative reaction to anti-H because the cells contain no H substance.
3. **D** A Bombay phenotype individual does not express A, B, or H antigens; therefore, anti-A, -B, and -H are formed. Because a Bombay phenotype individual has three antibodies, the only compatible blood must be from another Bombay phenotype donor.
4. **D** The acquired B phenomenon is only seen in group A persons.
5. **C** The patient is likely an A₂ with anti-A₁, which is causing reactivity in the crossmatching. A negative antibody screen rules out the possibility of an antibody to a high-frequency antigen, and two donor units incompatible rules out an antibody to a low-frequency antigen.

6. A patient's RBCs forward as group O, serum agglutinates B cells (4+) only. Your next step would be:

- A. Extend reverse typing for 15 minutes
- B. Perform an antibody screen, including room-temperature incubation
- C. Incubate washed RBCs with anti-A₁ and anti-A,B for 30 minutes at room temperature
- D. Test patient's RBCs with *Dolichos biflorus*

Blood bank/Apply principles of special procedures/RBCs/ABO discrepancy/3

7. Which typing results are most likely to occur when a patient has an acquired B antigen?

- A. Anti-A 4+, anti-B-3+, A₁ cells neg, B cells neg
- B. Anti-A 3+, anti-B neg, A₁ cells neg, B cells neg
- C. Anti-A 4+, anti-B 1+, A₁ cells neg, B cells 4+
- D. Anti-A 4+, anti-B 4+, A₁ cells 2+, B cells neg

Blood bank/Evaluate laboratory data to recognize problems/ABO discrepancy/2

8. Which blood group has the least amount of H antigen?

- A. A₁B
- B. A₂
- C. B
- D. A₁

Blood bank/Apply knowledge of fundamental biological principles/ABO/1

9. What type RBCs can be transfused to an A₂ person with anti-A₁?

- A. A only
- B. A or O
- C. B
- D. AB

Blood bank/Apply knowledge of fundamental biological principles/ABO/3

10. What should be done if all forward and reverse ABO results as well as the autocontrol are positive?

- A.** Wash the cells with warm saline, and autoadsorb the serum at 4°C
- B.** Retype the sample using a different lot number of reagents
- C.** Use polyclonal typing reagents
- D.** Report the sample as group AB

Blood bank/Evaluate laboratory and clinical data to specify additional tests/RBCs/ABO discrepancy/3

11. What should be done if all forward and reverse ABO results are negative?

- A.** Perform additional testing, such as typing with anti-A₁ lectin and anti-A,B
- B.** Incubate at 22°C or 4°C to enhance weak expression
- C.** Repeat the test with new reagents
- D.** Run an antibody identification panel

Blood bank/Evaluate laboratory and clinical data to specify additional tests/RBCs/ABO discrepancy/3

Answers to Questions 6–11

- 6. C** The strong 4+ reaction in reverse grouping suggests the discrepancy is in forward grouping. Incubating washed RBCs at room temperature with anti-A and anti-A,B will enhance reactions.
- 7. C** In forward typing, a 1+ reaction with anti-B is suspicious because of the weak reaction and the normal reverse grouping that appears to be group A. This may be indicative of an acquired antigen. In the case of an acquired B antigen, the reverse grouping is the same for a group A person. Choice A is indicative of group AB; choice B is indicative of a group A who may be immunocompromised. Choice D may be caused by a mistyping or an antibody against antigens on reverse cells.
- 8. A** The A₁B blood group has the least amount of H antigen. This occurs because both A and B epitopes are present on RBCs, compromising the availability of H epitopes. A₁B cells will yield weak reactions with anti-H lectin.
- 9. B** A person who is in need of RBC transfusion and is an A₂ with anti-A₁ can be transfused with A or O cells because the anti-A₁ is typically only reactive at room temperature.
- 10. A** These results point to a cold autoantibody. Washing the cells with warm saline may elute the autoantibody, allowing a valid forward type to be performed. The serum should be adsorbed using washed cells until the autocontrol is negative. Then, the adsorbed serum should be used for reverse typing.
- 11. B** All negative results may be caused by weakened antigens or antibodies. Room temperature or lower incubation temperature may enhance expression of weakened antigens or antibodies.

12. N-acetyl-D-galactosamine is the immunodominant carbohydrate that reacts with:

- A.** *Arachis hypogaea*

- B. *Salvia sclarea*
- C. *Dolichos biflorus*
- D. *Ulex europeaus*

Blood bank/Apply knowledge of fundamental biological principles/ABO/2

- 13. A stem cell transplant recipient was retyped when she was transferred from another hospital. What is the most likely cause of the following results?**

Patient cells:	Anti-A, neg	Anti-B, 4+
Patient serum:	A ₁ cells, neg	B cells, neg

- A. Viral infection
- B. Alloantibodies
- C. Immunodeficiency
- D. Autoimmune hemolytic anemia

Blood bank/Evaluate laboratory data to recognize health and disease states/ABO discrepancy/3

- 14. What reaction would be the same for an A₁ and an A₂ person?**

- A. Positive reaction with anti-A₁ lectin
- B. Positive reaction with A₁ cells
- C. Equal reaction with anti-H
- D. Positive reaction with anti-A,B

Blood bank/Evaluate laboratory data to make identifications/ABO discrepancy/2

- 15. A female patient at 28 weeks' pregnancy yields the following results:**

Patient cells:	Anti-A, 3+	Anti-B, 4+
Patient serum:	A ₁ cells, neg	B cells, 1+ O cells, 1+

Which of the following could be causing the ABO discrepancy?

- A. Hypogammaglobulinemia
- B. Alloantibody in patient serum
- C. Acquired B
- D. Weak subgroup

Blood bank/Evaluate laboratory data to make identifications/ABO discrepancy/3

- 16. Which condition would most likely be responsible for the following typing results?**

Patient cells:	Anti-A, neg	Anti-B, neg
Patient serum:	A ₁ cells, neg	B cells, 4+

- A. Immunodeficiency
- B. Masking of antigens by the presence of massive amounts of antibody
- C. Weak or excessive antigen(s)
- D. Impossible to determine

Blood bank/Apply principles of basic laboratory procedures/ABO discrepancy/3

- 17. Which of the following results is most likely discrepant?**

Anti-A, neg	Anti-B, 4+
A ₁ cells, neg	B cells, neg

- A. Negative B cells
- B. Positive reaction with anti-B
- C. Negative A₁ cells
- D. No problem with this typing

Blood bank/Evaluate laboratory data to make identifications/ABO discrepancy/3

Answers to Questions 12–17

12. **C** The immunodominant sugar N-acetyl-galactosamine confers A antigen specificity when present at the terminus of the type 2 precursor chain on the RBC membrane. Therefore, its presence would cause RBCs to react with anti-A₁ lectin, *Dolichos biflorus*.
13. **C** The transplant recipient is probably taking immunosuppressive medication to increase graft survival. This can contribute to the loss of normal blood group antibodies as well as other types of antibodies.
14. **D** Anti-A,B should react positively with group A or B and any subgroup of A or B (with exception of A_m). An A₁ (not A₂) would react with anti-A₁ lectin; only an A₂ person with anti-A₁ would give a positive reaction with A₁ cells; an A₂ would react more strongly with anti-H than A₁.
15. **B** The patient is most likely an AB person who has formed a cold-reacting alloantibody reacting with B cells and O cells. An identification panel should be performed. An acquired B person or someone with hypogammaglobulinemia should not make antibody that would agglutinate O cells.
16. **C** Excessive A substance, as may be found in some types of tumors, may be neutralizing anti-A. Weak A subgroups may fail to react with anti-A and require additional testing techniques (e.g., room-temperature incubation) before their expression is apparent.
17. **C** The reverse typing should agree with the forward typing in this result. The 4+ reaction with anti-B indicates group B. A positive reaction is expected with A₁ cells in the reverse group.

18. A 61-year-old male with a history of multiple myeloma underwent stem cell transplantation 3 years ago. The donor was O positive, and the recipient was B positive. The patient is admitted to a community hospital for fatigue and nausea. Typing results reveal the following:

Anti-A = 0
 Anti-B = 0
 Anti-A,B = 0
 Anti-D = 4+
 A₁ cells = 4+
 B cells = 0

How would you report this type?

- A. O positive

- B.** B positive
- C.** A positive
- D.** Undetermined

Blood bank/Evaluate laboratory data to recognize problems/ABO discrepancy/3

Answer to Question 18

- 18. D** In a transplantation scenario, there are no methods to employ to solve the discrepancy. The medical laboratory scientist must rely on the patient history of donor type and recipient type and the present serological picture. Giving a B-positive recipient an O-positive transplant constitutes a minor ABO mismatch. The forward type resembles the donor. The reverse type still resembles the recipient. The ABO type reported out does not fit a pattern resulting in an undetermined type.

4.3

Rh Blood Group System

1. **A complete Rh typing for antigens C, c, D, E, and e revealed negative results for C, D, and E. How is the individual designated?**

A. Rh positive
B. Rh negative
C. Positive for c and e
D. Impossible to determine

Blood bank/Apply knowledge of fundamental biological characteristics/Rh typing/1

2. **How is an individual with genotype *Dce/dce* classified?**

A. Rh positive
B. Rh negative
C. Rh_{null}
D. Total Rh

Blood bank/Apply knowledge of fundamental biological characteristics/Rh typing/2

3. **If a patient has a positive DAT, should you perform a weak D test on the cells?**

A. No, the cells are already coated with antibody
B. No, the cells are Rh_{null}
C. Yes, the immunoglobulin will not interfere with the test
D. Yes, Rh reagents are enhanced in protein media

Blood bank/Apply knowledge of fundamental biological characteristics/Rh typing/3

4. **Which donor unit is selected for a recipient with anti-c?**

A. r'r
B. R₀R₁
C. R₂r'
D. r'r^y

Blood bank/Apply knowledge of fundamental biological characteristics/Rh typing/3

5. **Which genotype usually shows the strongest reaction with anti-D?**

A. DCE/DCE
B. Dce/dCe
C. D-/D-
D. -CE/-ce

Blood bank/Apply knowledge of fundamental biological characteristics/Rh typing/1

Answers to Questions 1–5

1. **B** Rh positive refers to the presence of D antigen; Rh negative refers to the absence of D antigen. These designations are for D antigen only and do not involve other Rh antigens.

2. **A** This individual has D antigen and is classified as Rh positive. Any genotype containing D antigen will be considered Rh positive.
3. **A** If a person has a positive DAT, the RBCs are coated with immunoglobulin. If a test for weak D were performed, the test would yield positive results independent of the presence or absence of D antigen on the RBCs.
4. **D** The designation r' is dCe and r^y is dCE , neither of which contains c antigen. The other three Rh types contain c antigen and could not be used in transfusion for a person with anti-c.
5. **C** The phenotype that results from $D-/D-$ is classified as enhanced D because it shows a stronger reaction than expected with anti-D. Such cells have a greater amount of D antigen than normal. This is thought to result from a larger quantity of precursors being available to the D genes because there is no competition from other Rh genes.

6. Why is testing for Rh antigens and antibodies different from ABO testing?

- A.** ABO reactions are primarily caused by IgM antibodies and usually occur at room temperature; Rh antibodies are IgG and agglutination usually requires a 37°C incubation and enhancement media
- B.** ABO antigens are attached to receptors on the outside of the RBC and do not require any special enhancement for testing; Rh antigens are loosely attached to the RBC membrane and require enhancement for detection
- C.** Both ABO and Rh antigens and antibodies have similar structures, but Rh antibodies are configured so that special techniques are needed to facilitate binding to Rh antigens
- D.** There is no difference in ABO and Rh testing; both may be conducted at room temperature with no special enhancement needed for reaction

Blood bank/Apply knowledge of fundamental biological characteristics/Rh system/1

7. Testing reveals a weak D that reacts 1+ after indirect antiglobulin testing (IAT). How is this result classified?

- A.** Rh-positive
- B.** Rh-negative, Du positive
- C.** Rh-negative
- D.** Rh-positive, Du positive

Blood bank/Apply knowledge of standard operating procedures/Components/Rh label/2

8. What is one possible genotype for a patient who develops anti-C antibody?

- A.** $R1r$
- B.** R^1R^1
- C.** $r'r$
- D.** rr

Blood bank/Apply knowledge of fundamental biological characteristics/Rh typing/2

9. A patient developed a combination of Rh antibodies: anti-C, anti-E, and anti-D. Can compatible blood be found for this patient?

- A.** It is almost impossible to find blood lacking C, E, and D antigens
- B.** rr blood could be used without causing a problem
- C.** R_0R_0 may be used because it lacks all three antigens
- D.** Although rare, $r^y r$ blood may be obtained from close relatives of the patient

Blood bank/Apply knowledge of fundamental biological characteristics/Rh antibodies/1

10. A patient tests positive for weak D but also appears to have anti-D in his serum. What may be the problem?

- A. Mixup of samples or testing error
- B. Most weak D individuals make anti-D
- C. The problem could be caused by a disease state
- D. A D mosaic may make antibodies to missing antigen parts

Blood bank/Apply knowledge to identify sources of error/Rh antibodies/2

11. Which offspring is *not* possible from a mother who is R^1R^2 and a father who is R^1r ?

- A. DcE/DcE
- B. Dce/DcE
- C. DcE/DCe
- D. Dce/dce

Blood bank/Evaluate laboratory data to verify test results/Rh system/Paternity testing/2

Answers to Questions 6–11

- 6. **A** Detection of ABO and Rh antigens and antibodies requires different reaction conditions. ABO antibodies are naturally occurring IgM molecules and react best at room temperature. Rh antibodies are generally immune IgG molecules that result from transfusion or pregnancy. Detection may require 37°C incubation and/or enhancement techniques.
- 7. **A** Blood tested for weak D that shows 1+ reaction after IAT is classified as Rh positive. The weak D designation is not noted in the reporting of the result.
- 8. **D** Only rr (dce/dce) does not contain C antigen. A person will form alloantibodies only to the antigens he or she lacks.
- 9. **B** The genotype rr (dce/dce) lacks D, C, and E antigens and would be suitable for an individual who has developed antibodies to all three antigens. This is the most common Rh-negative genotype and is found in nearly 14% of blood donors who are white.
- 10. **D** The D antigen comprises different parts designated as a mosaic. If an individual lacks parts of the antigen, he or she may make antibodies to the missing parts if exposed to the whole D antigen.
- 11. **A** DcE/DcE (R^2R^2) is not possible because R^2 can be inherited only from the mother and is not present in the father.

12. Which weak D phenotypes necessitate Rh immune globulin (RhIg) prophylaxis?

- A. 1
- B. 2
- C. 3
- D. None of the above

Blood bank/Apply knowledge of biological characteristics/Rh testing/3

13. What antibodies could an R_1R_1 make if exposed to R_2R_2 blood?

- A. Anti-e and anti-C
- B. Anti-E and anti-c
- C. Anti-E and anti-C
- D. Anti-e and anti-c

Blood bank/Apply knowledge of fundamental biological characteristics/Rh antibodies/2

14. What does the genotype —/— represent in the Rh system?

- A. Rh negative
- B. D mosaic
- C. Rh_{null}
- D. Total Rh

Blood bank/Evaluate laboratory data to make identifications/Rh system/Rh antigens/2

15. What techniques are necessary for weak D testing?

- A. Saline + 22°C incubation
- B. Albumin or LISS + 37°C incubation
- C. Saline + 37°C incubation
- D. 37°C incubation + IAT

Blood bank/Apply knowledge of basic laboratory procedures/Rh system/2

16. A patient types as AB and appears to be Rh positive on slide typing. What additional tests should be performed for tube typing?

- A. Rh negative control
- B. Direct antiglobulin test
- C. Low-protein Rh antisera
- D. No additional testing is needed

Blood bank/Evaluate laboratory data to verify test results/Rh system/2

17. According to the Wiener nomenclature and/or genetic theory of Rh inheritance:

- A. There are three closely linked loci, each with a primary set of allelic genes
- B. The alleles are named R^1 , R^2 , R^0 , r , r' , r'' , R^z , and r^y
- C. There are multiple alleles at a single complex locus that determine each Rh antigen
- D. The antigens are named D, C, E, c, and e

Blood bank/Apply knowledge of fundamental biological principles/Rh system/2

18. The Wiener nomenclature for the E antigen is:

- A. hr'
- B. hrv'
- C. rh''
- D. Rh₀

Blood bank/Apply knowledge of fundamental biological principles/Rh typing/1

Answers to Questions 12–18

- 12. D** Weak D phenotypes 1, 2, and 3 (as well as 4) are not associated with formation of anti-D negating RhIg prophylaxis. Weak D phenotypes 5, 11, 15, 19, and 20 would require RhIg prophylaxis.
- 13. B** The R_1R_1 (DcE/DcE) individual does not have the E or c antigen, and could make anti-E and anti-c antibodies when exposed to R_2R_2 cells (DcE/DcE).

14. **C** A person who is Rh_{null} shows no Rh antigens on his or her RBCs. Loss of Rh antigens is very unlikely to happen because Rh antigens are integral parts of the RBC membrane. The Rh_{null} phenotype can result from either genetic suppression of the Rh genes or inheritance of amorphic genes at the Rh locus.
15. **D** Weak D testing requires both 37°C incubation and the IAT procedure. Anti-D is an IgG antibody, and attachment of the D antigen is optimized at warmer temperatures. AHG in the IAT phase facilitates lattice formation by binding to the antigen–antibody complexes.
16. **A** An Rh-negative control (patient cells in saline or 6% albumin) should be run if a sample appears to be AB positive. The ABO test serves as the Rh control for other ABO types.
17. **C** Wiener proposed a single-locus theory for Rh, with multiple alleles determining surface molecules that embody numerous antigens.
18. **C** The Wiener designation for the E antigen is rh". The Wiener designation hr' denotes c, hr" denotes e, and Rh₀ is D.

19. A physician orders 2 units of leukocyte-reduced RBCs. The patient is a 55-year-old male with anemia. He types as an AB negative, and his antibody screen is negative. There is only 1 unit of AB negative in inventory. What is the next blood type that should be given?

- A. AB positive (patient is male)
- B. A negative
- C. B negative
- D. O negative

Blood bank/Evaluate sources of errors/Rh systems/3

20. Which technology may report an Rh-weak D positive as Rh negative?

- A. Gel system
- B. Solid phase
- C. Tube testing
- D. None of these options

Blood bank/Apply knowledge of laboratory procedures/Rh system/2

Answers to Questions 19–20

19. **B** Although giving Rh-positive RBCs to an Rh-negative patient would not harm the patient in this case, because he is male, giving A negative would be the first choice. You should not expose a patient to the D antigen, if possible, and the residual anti-B in a unit of A-negative packed cells is less immunogenic than giving B or O RBCs.
20. **A** The Gel system cannot detect a weak D phenotype because there is no washing phase in the Gel system.

4.4

Testing for Antibodies

1. **A patient has the Lewis phenotype Le(a-b-). An antibody panel reveals the presence of anti-Le^a. Another patient with the phenotype Le(a-b+) has a positive antibody screen; however, a panel reveals no conclusive antibody. Should anti-Le^a be considered a possibility for the patient with the Le(a-b+) phenotype?**
 - A. Anti-Le^a should be considered as a possible antibody
 - B. Anti-Le^a may be a possible antibody, but further studies are needed
 - C. Anti-Le^a is not a likely antibody because even Le^b individuals secrete some Le^a
 - D. Anti-Le^a may be found in saliva but not detectable in serum

Blood bank/Apply knowledge of fundamental biological characteristics/Blood groups/2
2. **A medical laboratory scientist (MLS) is having great difficulty resolving an antibody mixture. One of the antibodies is anti-Le^a. This antibody is not clinically significant in this situation, but it needs to be removed to reveal the possible presence of an underlying antibody of clinical significance. What can be done?**
 - A. Perform an enzyme panel
 - B. Neutralize the serum with saliva
 - C. Neutralize the serum with hydatid cyst fluid
 - D. Use dithiothreitol (DTT) to treat the panel cells

Blood bank/Apply knowledge of fundamental biological characteristics/Blood groups/3
3. **What type of blood should be given to an individual who has anti-Le^b that reacts 1+ at the IAT phase?**
 - A. Blood that is negative for Le^b antigen
 - B. Blood that is negative for both Le^a and Le^b antigens
 - C. Blood that is positive for Le^b antigen
 - D. Lewis antibodies are not clinically significant, so any type of blood may be given

Blood bank/Apply knowledge of fundamental biological characteristics/Blood group antibodies/3
4. **Which of the following statements is true concerning the MN genotype?**
 - A. Antigens are destroyed using bleach-treated cells
 - B. Dosage effect may be seen for both M and N antigens
 - C. Both M and N antigens are impossible to detect because of cross-interference
 - D. MN is a rare phenotype seldom found in routine antigen typing

Blood bank/Apply knowledge of fundamental biological characteristics/Blood groups/2

Answers to Questions 1–4

1. **C** Anti-Le^a is produced primarily by persons with the Le(a-b-) phenotype because Le(a-b+) persons still have some Le^a antigen present in saliva. Although Le^a is not present on their RBCs, Le(a-b+) persons do not form anti-Le^a.

2. **B** Saliva from an individual with the *Le* gene contains the Le^a antigen. This combines with anti- Le^a , neutralizing the antibody. Panel cells treated with DTT (0.2M) lose reactivity with anti-K and other antibodies, but not anti- Le^a . Hydatid cyst fluid neutralizes anti- P_1 .
 3. **A** Lewis antibodies are generally not considered clinically significant unless they react at 37°C or at the IAT phase. The antibody must be honored in this scenario due to its reactivity at the IAT phase of testing.
 4. **B** *Dosage effect* is the term used to describe the phenomenon of an antibody that reacts more strongly with homozygous cells than with heterozygous cells. Dosage effect is a characteristic of the genotype *MN* because the M and N antigens are both present on the same cell. This causes a weaker reaction than seen with RBCs of either the *MM* or *NN* genotype, which carry a greater amount of the corresponding antigen.
5. **Anti-M is sometimes found with reactivity detected at the immediate spin (IS) phase that persists in strength to the IAT phase. What is the main testing problem with a strong anti-M?**
- A. Anti-M may not allow detection of a clinically significant antibody
 - B. Compatible blood may not be found for the patient with a strongly reacting anti-M
 - C. Anti-M cannot be removed from serum
 - D. Anti-M may react with the patient's own cells, causing a positive autocontrol
- Blood bank/Apply knowledge of fundamental biological characteristics/Blood groups/2*
6. **A patient is suspected of having paroxysmal cold hemoglobinuria (PCH). Which pattern of reactivity is characteristic of the Donath-Landsteiner antibody, which causes this condition?**
- A. The antibody attaches to RBCs at 4°C and causes hemolysis at 37°C
 - B. The antibody attaches to RBCs at 37°C and causes agglutination at the IAT phase
 - C. The antibody attaches to RBCs at 22°C and causes hemolysis at 37°C
 - D. The antibody attaches to RBCs and causes agglutination at the IAT phase
- Blood bank/Apply knowledge of fundamental biological characteristics/Blood group antibodies/1*
7. **How can interfering anti- P_1 antibody be removed from a mixture of antibodies?**
- A. Neutralization with saliva
 - B. Agglutination with human milk
 - C. Combination with urine
 - D. Neutralization with hydatid cyst fluid
- Blood bank/Apply principles of special procedures/Blood group antibodies/1*
8. **Which antibody is frequently seen in patients with warm autoimmune hemolytic anemia (WAIHA)?**
- A. Anti- Jk^a
 - B. Anti-e
 - C. Anti-K
 - D. Anti-Fy^b
- Blood bank/Apply knowledge of fundamental biological characteristics/Blood group antibodies/2*

9. A patient's antibody shows strong reactions in all test phases. All screen and panel cells are positive. Serum is then tested with a cord blood cell, and the reaction is negative. What antibody is suspected?

A. Anti-I
B. Anti-i
C. Anti-H
D. Anti-p

Blood bank/Apply principles of special procedures/Antibody ID/2

10. Which group of antibodies is commonly found as cold agglutinins?

A. Anti-K, anti-k, anti-Js^b
B. Anti-D, anti-e, anti-C
C. Anti-M, anti-N
D. Anti-Fy^a, anti-Fy^b

Blood bank/Apply knowledge of fundamental biological characteristics/Blood group antibodies/1

11. Which of the following antibodies characteristically gives a refractile mixed-field appearance?

A. Anti-K
B. Anti-Di^a
C. Anti-Sd^a
D. Anti-s

Blood bank/Apply knowledge of fundamental biological characteristics/Blood group antibodies/1

Answers to Questions 5–11

5. **A** Although anti-M may not be clinically significant, a strongly reacting anti-M that persists through to the IAT phase may interfere with detection of a clinically significant antibody that reacts only at IAT.
6. **A** The Donath-Landsteiner antibody has anti-P specificity with biphasic activity. The antibody attaches to RBCs at 4°C and then causes the RBCs to hemolyze when warmed to 37°C.
7. **D** Hydatid cyst fluid contains P₁ substance, which can neutralize anti-P₁ antibody.
8. **B** Anti-e is frequently implicated in cases of WAIHA.
The corresponding antigen is characterized as high frequency in the Rh system and can mask the presence of other alloantibodies.
9. **A** Adult cells contain mostly I antigen, and anti-I would react with all adult cells found on screen or panel cells. Cord blood cells, however, contain mostly i antigen and would test negative or only weakly positive with anti-I.
10. **C** Antibodies to the M and N antigens are IgM antibodies commonly found as cold agglutinins.
11. **C** Anti-Sd^a characteristically gives a refractile mixed-field agglutination reaction in the IAT phase. The refractile characteristic is more evident under the microscope.

12. What does the 3+3 rule ascertain?

- A. An antibody is ruled in
- B. An antibody is ruled out
- C. 95% confidence that the correct antibody has been identified
- D. 95% confidence that the correct antibody has not been identified

Blood bank/Apply principles of basic laboratory procedures/Antibody ID/1

13. The k (Cellano) antigen is a high-frequency antigen and is found on most RBCs. How often would one expect to find the corresponding antibody?

- A. Often, because it is a high-frequency antibody
- B. Rarely, because most individuals have the antigen and therefore would not develop the antibody
- C. Depends on the population, because certain racial and ethnic groups show a higher frequency of anti-k
- D. Impossible to determine without consulting regional blood group antigen charts

Blood bank/Calculate/Hemotherapy/1

14. Which procedure would help to distinguish between anti-e and anti-Fy^a in an antibody mixture?

- A. Lowering the pH of test serum
- B. Running an enzyme panel
- C. Using a thiol reagent
- D. Running an LISS panel

Blood bank/Apply principles of special procedures/Antibody ID/2

15. Which characteristics are true of all three of the following antibodies: anti-Fy^a, anti-Jk^a, and anti-K?

- A. Detected at the IAT phase; may cause hemolytic disease of the fetus and newborn (HDFN) and hemolytic transfusion reactions
- B. Not detected with enzyme-treated cells
- C. Requires the IAT technique for detection; usually not associated with HDFN
- D. Enhanced reactivity with enzyme-treated cells; may cause severe hemolytic transfusion reactions

Blood bank/Apply principles of special procedures/Antibody ID/2

16. A patient is admitted to the hospital. Medical records indicate that the patient has a history of anti-Jk^a. When you performed the type and screen, the type was O positive, and the screen was negative. You should:

- A. Crossmatch using units negative for Jk^a antigen
- B. Crossmatch random units, since the antibody is not demonstrating
- C. Request a new sample
- D. Repeat the screen with enzyme-treated screening cells

Blood bank/Apply principles of basic laboratory procedures/Antibody ID/3

Answers to Questions 12–16

- 12. C** The 3+3 rule ascertains correct identification of antibody at a confidence level of 95%. For this level to be met, reagent RBCs are found containing target antigen to suspected antibody that react in test phase; likewise, reagent RBCs devoid of antigen

will not react in test phase.

- 13. **B** k antigen is found with a frequency of 99.8%; therefore, k-negative individuals are rare. Because of this, the occurrence of anti-k is also rare.
- 14. **B** Enzyme-treated cells will not react with Duffy antibodies. Rh antibodies react more strongly with enzyme-treated RBCs. An enzyme panel, therefore, would enhance reactivity of anti-e and destroy reactivity to anti-Fy^a.
- 15. **A** Anti-Fy^a, anti-Jk^a, and anti-K are usually detected at IAT and all may cause HDFN and transfusion reactions that may be hemolytic. Reactivity with anti-Fy^a is lost with enzyme-treated RBCs, but reactivity with anti-Jk^a is enhanced with enzyme-treated cells. Reactivity with anti-K is unaffected by enzyme-treated cells.
- 16. **A** The Kidd antibodies are notorious for disappearing from serum, yielding a negative result for the antibody screen. If a patient has a history of a Kidd antibody, blood must be crossmatched using antigen-negative units. If the patient is transfused with the corresponding antigen, an anamnestic response may occur with a subsequent hemolytic transfusion reaction.

17. An MLS performs an antibody study and finds 1+ and weak positive reactions for several of the panel cells. The reactions do not fit a pattern. Several selected panels and a patient phenotype do not reveal any additional information. Serum is diluted and retested, but the same reactions persist. What type of antibody may be causing these results?

- A. Antibody to a high-frequency antigen
- B. Antibody to a low-frequency antigen
- C. High titer low avidity (HTLA)
- D. Anti-human leukocyte antigen (anti-HLA)

Blood bank/Evaluate laboratory data to make identifications/Antibody ID/3

18. An antibody is detected in a pregnant woman and is suspected of being the cause of fetal distress. The antibody reacts at the IAT phase but does not react with DTT-treated cells. This antibody causes in vitro hemolysis. What is the most likely antibody specificity?

- A. Anti-Le^a
- B. Anti-Lu^a
- C. Anti-Lu^b
- D. Anti-Xg^a

Blood bank/Evaluate laboratory data to make identifications/Antibody ID/3

19. What sample is best for detecting complement-dependent antibodies?

- A. Plasma stored at 4°C for no longer than 24 hours
- B. Serum stored at 4°C for no longer than 48 hours
- C. Either serum or plasma stored at 20°C to 24°C no longer than 6 hours
- D. Serum heated at 56°C for 30 minutes

Blood bank/Apply principles of basic laboratory procedures/Antibody ID/2

20. Which antibody would not be detected by group O screening cells?

- A. Anti-N
- B. Anti-A₁

C. Anti-Dj^a

D. Anti-k

Blood bank/Apply principles of special procedures/Antibody ID/1

21. Refer to Panel 1. Which antibody is most likely implicated?

A. Anti-Fy^b

B. Anti-Jk^b

C. Anti-e

D. Anti-c and anti-K

Blood bank/Apply principles of special procedures/Antibody ID/2

Answers to Questions 17–21

17. C HTLA antibodies may persist in reaction strength, even when diluted. These antibodies are directed against high-frequency antigens (e.g., Ch^a). They are not clinically significant but, when present, are responsible for a high incidence of incompatible crossmatches.

18. C Of the antibodies listed, only Lu^b is detected in the IAT phase, causes in vitro hemolysis, may cause HDFN, and does not react with DTT-treated cells.

19. B Serum stored at 4°C for no longer than 48 hours preserves complement activity. Plasma is inappropriate because most anticoagulants chelate calcium needed for activation of complement. Heating the serum to 56°C destroys complement.

20. B ABO antibodies are not detected by group O screening cells because O cells contain no A or B antigens.

21. B The pattern clearly fits that of anti-Jk^b, an antibody that usually reacts best at IAT. The weaker reactions are caused by the dosage effect found on cells that are heterozygous for the Jk^b antigen.

Panel 1																								
Cell	D	C	E	c	e	K	k	Kp ^a	Kp ^b	J ^s _a	J ^s _b	F ^y _a	F ^y _b	J ^k _a	J ^k _b	X ^g _a	L ^e _a	L ^e _b	S	s	M	N	P	
1	+	+	O	O	+	O	+	O	+	O	+	O	+	O	+	+	+	O	O	+	O	+	C	
2	+	+	O	O	+	+	+	O	+	O	+	+	+	+	+	+	+	O	+	+	+	O	+	+
3	+	O	+	+	O	O	+	O	+	O	+	+		+	+	O	O	+	+	+	+	+	+	+
4	O	+	O	+	+	O	+	O	+	O	+	+	+	+	+	+	+	O	+	O	+	+	+	C
5	O	O	+	+	+	+	+	O	+	O	+	+	+	+	O	O	O	+	O	+	+	+	+	+
6	O	O	O	O	+	O	+	O	+	O	+	+	+	+	+	+	+	O	+	+	+	O	+	+
7	O	O	O	+	+	+	+	O	+	O	+	O		+	O	+	O	+	+	O	+	O	+	+
8	O	O	O	+	+	O	+	O	+	O	+	O	+	O	+	+	O	O	+	+	+	O	+	+
9	+	+	O	O	+	O	+	O	+	O	+	+	+	+	+	+	+	O	O	+	+	+	+	+

10 + O O + + O + O + O + O O + + + O O O O + O +

22. Refer to Panel 2. Which antibody specificity is most likely present?

- A. Anti-S and anti-E
- B. Anti-E and anti-K
- C. Anti-Le^a and anti-Fy^b
- D. Anti-C and anti-K

Blood bank/Apply principles of special procedures/Antibody ID/3

23. On Panel 2, which of the following antibodies could *not* be ruled out?

- A. Anti-Jk^b
- B. Anti-C
- C. Anti-M
- D. Anti-Fy^b

Blood bank/Apply principles of special procedures/Antibody ID/3

24. On Panel 2, which cells are homozygous for C?

- A. 1, 2, 3
- B. 1, 2, 9
- C. 3, 4, 7
- D. 7, 8, 10

Blood bank/Apply principles of special procedures/Antibody ID/2

Answers to Questions 22–24

- 22. D** The pattern fits anti-C at 37°C, which becomes stronger at the IAT phase. The additional antibody is anti-K, which appears only at the IAT phase.
- 23. B** To rule out an antibody, there should be a homozygous cell with the corresponding antigen that fails to react with serum. Of the choices, anti-C was not ruled out on Panel 2. To rule this antibody out, a cell that is homozygous for C and negative for K (the other probable antibody) would be run against patient serum. A positive reaction supports the presence of anti-C, whereas a negative reaction would rule out anti-C.
- 24. B** On panel cells 1, 2, and 9, C antigen is present and c antigen is absent, rendering the cells homozygous for C.

Panel 2																						
Cell	D	C	E	c	e	K	k	Kp ^a	Kp ^b	Js ^a	Js ^b	Fy ^a	Fy ^b	Jk ^a	Jk ^b	Xg ^a	Le ^a	Le ^b	S	s	M	N
1	+	+	O	O	+	O	+	O	+	O	+	O	+	O	+	+	+	O	O	+	O	+
2	+	+	O	O	+	+	+	O	+	O	+	+	+	+	+	+	O	+	+	+	O	+
3	+	O	+	+	O	O	+	O	+	O	+	+	O	+	+	O	O	+	+	+	+	+
4	O	+	O	+	+	O	+	O	+	O	+	+	+	+	+	+	O	+	O	+	+	+
5	O	O	+	+	+	+	+	O	+	O	+	+	+	+	O	O	O	+	O	+	+	+
6	O	O	O	O	+	O	+	O	+	O	+	+	+	+	+	+	+	O	+	+	+	O

9	0	0	0	+	+	0	+	0	+	0	+	+	+	+	+	+	+	0	0	+	+	+	+
10	+	+	0	+	+	+	+	0	+	0	+	+	+	0	+	+	0	+	+	+	+	+	+

4.5

Compatibility Testing

1. **SITUATION: An emergency trauma patient requires transfusion. Six units of blood are ordered *stat* (immediately). There is no time to draw a patient sample. O-negative blood is issued. When will compatibility testing be performed?**
 - A. Compatibility testing must be performed before blood is issued
 - B. Compatibility testing will be performed when a patient sample is available
 - C. Compatibility testing may be performed immediately by using donor serum
 - D. Compatibility testing is not necessary when blood is issued in emergency situations

Blood bank/Apply knowledge of laboratory operations/Crossmatch/3
2. **How would autoantibodies affect compatibility testing?**
 - A. No effect
 - B. The DAT result would be positive
 - C. ABO, Rh, antibody screen, and crossmatching may show abnormal results
 - D. Results would depend on the specificity of autoantibody

Blood bank/Evaluate laboratory data to make identifications/Antibody ID/3
3. **An antibody screen is reactive at the IAT phase of testing with all three cells of a three-cell screen, and the autocontrol is negative. What is a possible explanation for these results?**
 - A. A cold alloantibody
 - B. High-frequency alloantibody or a mixture of alloantibodies
 - C. A warm autoantibody
 - D. A cold and warm alloantibody

Blood bank/Evaluate laboratory data to make identifications/Antibody ID/3
4. **What does a minor crossmatch consist of?**
 - A. Recipient plasma and recipient RBCs
 - B. Recipient plasma and donor RBCs
 - C. Recipient RBCs and donor plasma
 - D. Donor plasma and donor RBCs

Blood bank/Apply knowledge of laboratory operations/Crossmatch/1
5. **Can crossmatching be performed on October 14 using a patient sample drawn on October 12?**
 - A. Yes, a new sample would not be needed
 - B. Yes, but only if the previous sample has no alloantibodies
 - C. No, a new sample is needed because the 2-day limit has expired
 - D. No, a new sample is needed for each testing

Blood bank/Apply knowledge of standard operating procedures/Crossmatch/2

1. **B** When patient serum is available, it will be crossmatched with donor cells. Patient serum might contain antibodies against antigens on donor cells that may destroy donor cells. If an incompatibility is discovered, the problem will be reported immediately to the patient's physician.
2. **C** Autoantibodies may cause positive reactions with screening cells, panel cells, donor cells, and patient cells. The DAT result will be positive; however, DAT is not included in compatibility testing.
3. **B** High-frequency alloantibodies or a mixture of alloantibodies may cause all three screening cells to be positive. A negative autocontrol would rule out autoantibodies.
4. **C** A minor crossmatch consists of recipient RBCs and donor serum or plasma.
5. **A** Compatibility testing may be performed on a patient sample within 3 days of the scheduled transfusion; however, if the patient is pregnant or was transfused within 3 months, the sample must be less than 3 days old.

6. A type and screen was performed on a 32-year-old woman, and the patient was typed as AB negative. There are no AB-negative units in the blood bank. What should be done?

- A. Order AB-negative units from a blood supplier
- B. Check inventory of A-, B-, and O-negative units
- C. Ask the patient to make a preoperative autologous donation
- D. Nothing—the blood will probably not be used

Blood bank/Apply principles of basic laboratory procedures/Crossmatch/2

7. What ABO types may donate to any other ABO type?

- A. A negative, B negative, AB negative, O negative
- B. O negative
- C. AB negative
- D. AB negative, A negative, B negative

Blood bank/Apply knowledge of fundamental biological characteristics/Crossmatch/2

8. What type(s) of RBC(s) is (are) acceptable to transfuse to an O-negative patient?

- A. A negative, B negative, AB negative, or O negative
- B. O negative
- C. AB negative
- D. AB negative, A negative, B negative

Blood bank/Apply knowledge of fundamental biological characteristics/Crossmatch/2

9. An MLS removed 4 units of blood from the blood bank refrigerator and placed them on the counter. A clerk was waiting to take the units for transfusion. As she checked the paperwork, she noticed that one of the units was leaking onto the counter. What should she do?

- A. Issue the unit if the RBCs appear normal
- B. Reseal the unit
- C. Discard the unit
- D. Call the medical director and ask for an opinion

Blood bank/Apply knowledge of standard operating procedures/Crossmatch/3

10. A donor was found to contain anti-K using pilot tubes from the collection procedure. How would this affect the compatibility test?

- A. The AHG major crossmatch would be positive
- B. The IS major crossmatch would be positive
- C. The recipient's antibody screen would be positive for anti-K
- D. Compatibility testing would not be affected

Blood bank/Apply principles of basic laboratory procedures/Crossmatch/2

11. Which of the following is *not* a requirement for electronic crossmatching?

- A. The computer system contains logic to prevent assignment and release of ABO-incompatible blood
- B. There are concordant results of at least two determinations of the recipient's ABO type on record, one of which is from the current sample
- C. Critical elements of the system have been validated on site
- D. There are concordant results of at least one determination of the recipient's ABO type on file

Blood bank/Apply principles of basic laboratory procedures/Crossmatch/1

12. A patient showed positive results with screening cells and 4 donor units. The patient autocontrol was negative. What is the most likely antibody?

- A. Anti-H
- B. Anti-S
- C. Anti-Kpa
- D. Anti-k

Blood bank/Evaluate laboratory data to make identifications/Incompatible crossmatch/3

Answers to Questions 6–12

- 6. **B** An AB person is the universal recipient and may receive any blood type; because only a type and screen were ordered and blood may not be used, check inventory for A-, B-, and O-negative units.
- 7. **B** An O-negative individual has no A or B antigens and may donate RBCs to any other ABO type.
- 8. **B** An O-negative individual has both anti-A and anti-B and may receive only O-negative RBCs.
- 9. **C** Leaking may indicate a broken seal or a puncture, which indicates possible contamination of the unit, even if the RBCs appear normal. The unit should be discarded.
- 10. **D** Compatibility testing would not be affected if the donor has anti-K in his or her serum. This is because the major crossmatch uses recipient serum and not donor serum. Other tests, such as ABO, Rh, and antibody screen on the recipient, also would not be affected.
- 11. **D** ABO determinations must be concordant on at least two occasions, including the current sample.
- 12. **D** Anti-k (Cellano) is a high-frequency alloantibody that would react with screening cells and most donor units. The negative autocontrol rules out autoantibodies. Anti-H and anti-S are cold antibodies and anti-Kpa is a low-frequency alloantibody.

13. Screening cells and major crossmatch are positive on IS only, and the autocontrol is negative. Identify the problem.

- A. Cold alloantibody
- B. Cold autoantibody
- C. Abnormal protein
- D. Antibody mixture

Blood bank/Evaluate laboratory data to make identifications/Incompatible crossmatch/3

14. Six units are crossmatched. Five units are compatible, one unit is incompatible, and the recipient's antibody screen is negative. Identify the problem:

- A. Patient may have an alloantibody to a high-frequency antigen
- B. Patient may have an abnormal protein
- C. Donor unit may have a positive DAT
- D. Donor may have a high-frequency antigen

Blood bank/Evaluate laboratory data to make identifications/Incompatible crossmatch/3

15. An incompatible donor unit is found to have a positive result on DAT. What should be done with the donor unit?

- A. Discard the unit
- B. Antigen type the unit for high-frequency antigens
- C. Wash the donor cells and use the washed cells for testing
- D. Perform a panel on the incompatible unit

Blood bank/Apply principles of special procedures/Incompatible crossmatch/3

16. Screening cells, major crossmatch, and patient autocontrol are positive in all phases. Identify the problem.

- A. Specific cold alloantibody
- B. Specific cold autoantibody
- C. Abnormal protein or nonspecific autoantibody
- D. Cold and warm alloantibody mixture

Blood bank/Evaluate laboratory data to make identifications/Incompatible crossmatch/3

17. A panel study has revealed the presence of patient alloantibodies. What is the first step in a major crossmatch?

- A. Perform a DAT on patient cells and donor units
- B. Antigen type patient cells and any donor cells to be crossmatched
- C. Adsorb any antibodies from patient serum
- D. Obtain a different enhancement medium for testing

Blood bank/Apply principles of special procedures/Incompatible crossmatch/2

18. What is the disposition of a donor RBC unit that contains an antibody?

- A. The unit must be discarded
- B. Only the plasma may be used to make components
- C. The antibody must be adsorbed from the unit
- D. The unit may be labeled to indicate that it contains antibody and released into inventory

Blood bank/Apply knowledge of laboratory operations/Hemotherapy/Blood components/1

19. Given a situation where screening cells, major crossmatch, autocontrol, and DAT (anti-IgG) are all positive, what procedure should be performed next?

- A. Adsorption using rabbit stroma
- B. Antigen typing of patient cells

- C. Elution followed by a cell panel on the eluate
- D. Selected cell panel

Blood bank/Apply principles of special procedures/Incompatible crossmatch/3

Answers to Questions 13–19

- 13. **A** A cold alloantibody would show a reaction with screening cells and donor units only at the IS phase. The negative autocontrol rules out autoantibodies and abnormal protein.
- 14. **C** The incompatible donor unit may have an antibody coating the RBCs, or the patient may have an alloantibody to a low-frequency antigen. An alloantibody to a high-frequency antigen would agglutinate all units and screening cells.
- 15. **A** The incompatible unit may have RBCs coated with antibody and/or complement. If RBCs are sensitized, then some problem exists with the donor. Discard the unit.
- 16. **C** An abnormal protein or nonspecific autoantibody would cause antibody screen, crossmatch, and patient autocontrol to be positive. Alloantibodies would not cause a positive patient autocontrol.
- 17. **B** Antigen typing or phenotyping of the patient's cells confirms the antibody identification; antigen typing of donor cells helps ensure the crossmatch of compatible donor units.
- 18. **D** The unit may be used in the general blood inventory, if it is properly labeled and only cellular elements are used.
- 19. **C** A positive DAT using anti-IgG indicates that antibodies are coating the patient's cells. An eluate would be helpful to remove the antibody, followed by a cell panel to identify it.

20. A major crossmatch and screening cells are 2+ at the IS phase, 1+ at 37°C, and negative at the IAT phase. Identify the most likely problem.

- A. Combination of antibodies
- B. Cold alloantibody
- C. Rouleaux
- D. Test error

Blood bank/Evaluate laboratory data to make identifications/Incompatible crossmatch/3

21. What corrective action should be taken when rouleaux causes positive test results?

- A. Perform a saline replacement technique
- B. Perform an autoabsorption
- C. Run a panel
- D. Perform an elution

Blood bank/Apply principles of special procedures/Testing problems/3

22. All of the following are reasons for performing an adsorption, except:

- A. Separation of mixtures of antibodies
- B. Removal of interfering substances
- C. Confirmation of weak antigens on RBCs

D. Identification of antibodies causing a positive DAT

Blood bank/Apply principles of special procedures/Antibody identification/2

23. How long must a recipient sample be kept in the blood bank after compatibility testing?

A. 3 days

B. 5 days

C. 7 days

D. 10 days

Blood bank/Apply principles of basic laboratory procedures/Compatibility/1

24. What is the crossmatching protocol for platelets and/or plasma?

A. Perform a reverse grouping on donor plasma

B. No testing is required

C. Perform a reverse grouping on recipient plasma

D. Platelets must be HLA compatible

Blood bank/Apply principles of basic laboratory procedures/Compatibility/2

25. What are the compatibility requirements for an autologous unit?

A. ABO and Rh typing

B. Type and screen

C. Major crossmatch

D. All of these options

Blood bank/Apply principles of basic laboratory procedures/Compatibility/1

26. A patient is typed as AB positive. Two units of blood have been ordered by the physician. Currently, the inventory shows no AB units, 10 A-positive units, 1 A-negative unit, 5 B-positive units, and 20 O-positive units. Which should be set up for the major crossmatch?

A. A-positive units

B. O-positive units

C. B-positive units

D. Call another blood supplier for type-specific blood

Blood bank/Apply principles of basic laboratory procedures/Compatibility/2

27. Which of the following comprises an abbreviated crossmatch?

A. ABO, Rh, and antibody screen

B. ABO, Rh, antibody screen, IS crossmatch

C. Type and screen

D. ABO, Rh, IS crossmatch

Blood bank/Apply principles of basic laboratory procedures/Crossmatch/2

Answers to Questions 20–27

20. B The reaction pattern fits that of a cold antibody reacting at the IS phase and of sufficient titer to persist at 37°C incubation. The reactions disappear in the IAT phase.

21. A Rouleaux may be dispersed or lessened by using the saline replacement technique. This involves recentrifuging the tube, then withdrawing serum and replacing it with two drops of saline. The tube is respun and examined for hemolysis.

22. D Antibodies causing a positive DAT would be coating RBCs and would require

elution, not adsorption, to identify them.

- 23. **C** According to the American Association of Blood Banks (AABB) standards, the recipient sample must be kept for 7 days after compatibility testing.
- 24. **B** For transfusion of platelets and plasma, there is no required protocol for crossmatching.
- 25. **A** The only requirement for transfusing an autologous unit is a check of the ABO and Rh types.
- 26. **A** The type chosen should be A-positive RBC units. Although all choices would be compatible, the first choice should be A-positive because this unit will contain residual plasma anti-B. Anti-B is less immunogenic than anti-A, which would be present, albeit in small amounts, in B-positive and O-positive units.
- 27. **B** The abbreviated crossmatch usually consists of a type and screen and an IS crossmatch.

28. When may IS crossmatching be performed?

- A.** When a patient is being massively transfused
- B.** When there is no history of antibodies and the current antibody screen is negative
- C.** When blood is being emergency released
- D.** When a patient has not been transfused in the past 3 months

Blood bank/Apply principles of basic laboratory procedures/Crossmatch/1

Answer to Question 28

- 28. **B** IS crossmatching may be performed when the patient has no history of antibodies and the current antibody screen is negative.

4.6

Transfusion Reactions

1. **A patient had a transfusion reaction to packed RBCs. The MLS began the laboratory investigation of the transfusion reaction by assembling pre- and post-transfusion specimens and all paperwork and computer printouts. What should the MLS do next?**
 - A. Perform a DAT on the post-transfusion sample
 - B. Check for clerical error(s)
 - C. Repeat ABO and Rh typing of patient and donor units
 - D. Perform an antibody screen on the post-transfusion sample

Blood bank/Apply knowledge of standard operating procedures/Transfusion reactions/2
2. **What is the pathophysiological cause surrounding anaphylactic and anaphylactoid reactions?**
 - A. Antibody in patient's serum is detected 3 to 7 days after transfusion and is attached to donor RBCs
 - B. Donor plasma has reagins (IgE or IgA) that combine with allergens in patient's plasma
 - C. Patient is deficient in IgE and develops IgE antibodies via sensitization as a result of transfusion or pregnancy
 - D. Patient is deficient in IgA and develops IgA antibodies via sensitization as a result of transfusion or pregnancy

Blood bank/Apply knowledge of fundamental biological principles/Transfusion reactions/1
3. **A patient has a hemolytic reaction to blood transfused 8 days ago. What is the most likely cause?**
 - A. Immediate, nonimmunologic, probably as a result of volume overload
 - B. Delayed immunologic, probably as a result of an antibody such as anti-Jk^a
 - C. Delayed nonimmunologic, probably as a result of iron overload
 - D. Immediate, immunologic, probably as a result of clerical error, ABO incompatibility

Blood bank/Apply knowledge of fundamental biological characteristics/Transfusion reactions/2
4. **What may be found in the serum of a person who is exhibiting signs of transfusion-related acute lung injury (TRALI)?**
 - A. RBC alloantibody
 - B. IgA antibody
 - C. Antileukocyte antibody
 - D. Allergen

Blood bank/Apply knowledge of fundamental biological characteristics/Transfusion reactions/1

Answers to Questions 1–4

1. **B** Over 90% of transfusion reactions are caused by some type of clerical error. The most

time-saving approach would be to check all paperwork before performing any laboratory testing.

2. **D** Anaphylactic or anaphylactoid reactions are the most severe form of allergic transfusion reaction and are associated with deficient or absent IgA in the patients, allowing the formation of anti-IgA. These patients must be transfused with washed cellular products where the plasma has been removed.
3. **B** A transfusion reaction that occurs several days after a transfusion of blood products is probably a delayed immunologic reaction as a result of an antibody formed against donor antigens. This is a classic example of a reaction caused by an antibody, such as anti-Jk^a.
4. **C** TRALI is associated with antibodies to human leukocyte antigens or neutrophil antigens, which react with the patient's granulocytes and cause acute respiratory insufficiency.

5. **Which type of transfusion reaction occurs in about 1% of all transfusions, results in a temperature rise of greater than 1°C above 37°C associated with blood component transfusion, and is not related to the patient's medical condition?**

- A. Immediate hemolytic
- B. Delayed hemolytic
- C. Febrile nonhemolytic reaction
- D. TRALI

Blood bank/Apply knowledge of fundamental biological characteristics/Transfusion reactions/2

6. **What would be the result of group A blood given to a group O patient?**

- A. Nonimmune transfusion reaction
- B. Immediate hemolytic transfusion reaction
- C. Delayed hemolytic transfusion reaction
- D. Febrile nonhemolytic transfusion reaction (FNHTR)

Blood bank/Apply knowledge of fundamental biological characteristics/Transfusion reactions/2

7. **Patient DB received 2 units of group A-positive RBCs 2 days ago. Two days later, he developed a fever and appeared jaundiced. His blood type was A positive. A transfusion reaction workup was ordered. There were no clerical errors detected. A post-transfusion specimen was collected and a DAT performed. The DAT result was positive with monospecific anti-IgG. The plasma was also hemolyzed. An antibody screen and panel studies revealed the presence of anti-Jk^b in the post-transfusion specimen. The antibody screen on the pretransfusion specimen was negative. Which of the following explains the positive DAT?**

- A. The donor cells had a positive DAT
- B. The donor cells were polyagglutinable
- C. The donor cells were likely positive for the Jk^b antigen
- D. The recipient cells were likely positive for the Jk^b antigen

Blood bank/Apply knowledge of fundamental biological characteristics/Transfusion reactions/3

8. **All of the following are part of the preliminary evaluation of a transfusion reaction,**

except:

- A.** Check pre- and post-transfusion samples for color of serum
- B.** Perform ABO and Rh recheck
- C.** Perform DAT on the post-transfusion sample
- D.** Perform a panel on pre- and post-transfusion samples

Blood bank/Apply knowledge of standard operating procedures/Transfusion reactions/1

- 9. A 68-year-old female diagnosed with neutropenia and inflammation of the left hand was typed as A positive and received 1 packed RBC unit. The antibody screen was negative, and crossmatch was compatible. During the transfusion, her pulse was 94, and blood pressure (BP) rose from 114/59 mm Hg to 132/64 mm Hg. Temperature rose from 37.1°C before transfusion to 37.8°C 60 minutes after starting transfusion and then to 38.3°C upon completion. A post-transfusion specimen yielded plasma that was neither hemolyzed nor icteric, and a negative DAT. Post-transfusion urinalysis showed 1+ blood and protein with 10 RBCs/high-power field (hpf) microscopically. The clerical check result was acceptable. What type of reaction most likely occurred as a result of transfusion?**

- A.** Allergic
- B.** Circulatory overload
- C.** Febrile nonhemolytic
- D.** Delayed hemolytic

Blood bank/Correlation of laboratory and clinical data/Transfusion reactions/3

Answers to Questions 5–9

- 5. C** FNHTR is defined as a rise in temperature of greater than 1°C above 37°C, associated with transfusion and unexplained by other causes. The patient has formed antibodies to HLA, which reacted with donor cells and resulted in release of pyrogens.
- 6. B** Group A blood given to a group O patient would cause an immediate hemolytic transfusion reaction because a group O patient has anti-A and anti-B antibodies and would destroy A cells.
- 7. C** This is an example of an anamnestic reaction, where the patient was most likely exposed to the Jk^b antigen at some point in his life, and upon re-exposure to the antigen, the antibody titer rose to detectable levels. This resulted in positive DAT and post-transfusion antibody screen results.
- 8. D** The preliminary evaluation of a transfusion reaction includes checking the color of serum and performing ABO and Rh checks and DAT on the post-transfusion sample. A panel would not be part of the preliminary workup.
- 9. C** The temperature rose from 37.1°C to 38.3°C. The DAT result was negative, and although blood was found on the urinalysis, microscopic RBCs were also found. Because intact RBCs are not caused by a transfusion reaction, the cause of hematuria was not likely transfusion related. A febrile nonhemolytic reaction is highly consistent with both symptoms and post-transfusion test results.

- 10. A 92-year-old male diagnosed with anemia and episodes of frequent falling was typed as**

B negative and transfused with 1 unit of packed RBCs, also B negative. He had not been recently transfused, and the antibody screen was negative. During the transfusion, his temperature rose from 36.2°C to 36.4°C, his pulse from 96 to 124, respirations from 18 to 20, and BP from 127/81 mm Hg to 174/83 mm Hg. He was transfused with 205 mL before a reaction was called by the transfusionist. The post-transfusion specimen DAT result was negative, and the clerical check result was acceptable. Urinalysis showed 1+ blood with 5 RBCs microscopically. Other symptoms included tachycardia and flushing. What reaction had most likely taken place?

- A. Febrile nonhemolytic**
- B. Acute hemolytic**
- C. Anaphylactic**
- D. Volume overload**

Blood bank/Correlation of laboratory and clinical data/Transfusion reaction/3

- 11. A 76-year-old female diagnosed with urosepsis was transfused with 2 units of packed RBCs. Her type was AB positive and she had a negative result on antibody screen. The units transfused were AB positive. Upon receiving the second unit, the patient became hypoxic with tachypnea. The clerical check result was acceptable, and the DAT was negative. She received 269 mL from the second unit before a reaction was called. Her temperature fell from 38°C to 36.4°C, her pulse increased from 72 to 90, and respirations rose from 35 to 41. Her BP was 110/70 mm Hg. The patient died approximately 12 hours after the reaction was called. What type of reaction was most likely present?**

- A. Febrile**
- B. Symptoms not related to transfusion**
- C. Allergic**
- D. TRALI**

Blood bank/Correlate laboratory and clinical data/Transfusion reactions/3

- 12. A 52-year-old male received 2 units of packed RBCs as an outpatient in the intravenous (IV) therapy unit. He had had a head trauma 20 years ago and was quadriplegic. He had recurrent pneumonia and hematuria as a result of removal of a Foley catheter. His blood type was A positive, with previously identified anti-Fy^a. There was an ABO discrepancy, in that reverse typing with reagent A₁ cells was positive. The MLS attributed this reaction to Fy^a antigen being present on the reagent A₁ cells. The patient also had a cold autoantibody. Two units of A-positive packed cells were crossmatched that were Fy^a negative and were compatible. One unit was transfused at 11:30 a.m. without incident. The second unit was transfused at 2:16 p.m. and stopped at 3:55 p.m. because of reddish brown-tinged urine found in his collection bag. A post-transfusion specimen yielded a positive DAT and plasma that was grossly hemolyzed. A prewarm crossmatch was incompatible in both the pre- and post-transfusion specimens. Anti-E and anti-c were present in the post-transfusion specimen. What reaction was most likely present?**

- A. Acute hemolytic**
- B. Febrile**
- C. Allergic**
- D. TRALI**

Blood bank/Correlate laboratory and clinical data/Transfusion reactions/3

Answers to Questions 10–12

- 10. D** The tachycardia, increased pulse, and volume transfused before a reaction was called are consistent with volume overload. The temperature change did not meet criteria for a febrile reaction, and evidence for a hemolytic reaction is lacking.
- 11. B** This case emphasizes the statistic that not all causes of death are related to transfusion. The temperature dropped ruling out a febrile reaction; there was no evidence of pulmonary edema or hypotension seen with TRALI (and plasma products are more associated with TRALI compared with RBCs); and there was no sign of hives or itching, which are often associated with an allergic reaction.
- 12. A** This case represents an acute hemolytic reaction where the patient had previous sensitization to E and c antigens. Given the history of anti-Fy^a, an assumption was made that anti-Fy^a was the cause of the positive reverse type with A₁ cells, even though this antibody does not react at the IS phase. This brings to light the importance of running a panel whenever the patient has a positive antibody screen, regardless of previous results. Hemoglobinuria, a positive DAT, and the hemolyzed post-transfusion specimen all are consistent with an acute hemolytic reaction.

13. An 82-year-old male was admitted for renal failure. His type was B positive, and his antibody screen was negative. Two units of RBCs were ordered. The first unit was transfused at 1:00 p.m. without incident. The second was started at 4:15 p.m. and stopped at 5:12 p.m. after the nurse observed that the patient had died. Vital signs had been taken at 4:30 p.m. with no abnormalities. A transfusion reaction was called and the blood unit, tubing, and paperwork sent to the blood bank. There were no clinical manifestations noted on the paperwork, and no post-transfusion specimen was sent to the blood bank. What type of reaction most likely occurred?

- A.** Cause not related to transfusion
- B.** Acute hemolytic reaction
- C.** Anaphylactic reaction
- D.** Volume overload

Blood bank/Correlate clinical and laboratory data/Transfusion reactions/3

Answer to Question 13

- 13. A** This example represents a situation where the pathologist was not provided with all information needed to interpret the reaction. There was no information on patient symptoms, the patient had received another unit of RBCs hours previously with no problems, and a postreaction specimen was not collected. Therefore, any serological abnormalities could not be identified. The U.S. Food and Drug Administration (FDA) recommends collecting a postmortem specimen if a reaction is called so that the investigation of transfusion reaction can be completed. In this case, the pathologist interpreted the reaction as not related to transfusion because no symptoms were documented.

4.7

Components

1. A male patient with cancer was admitted to the hospital with acute abdominal pain and a hemoglobin level of 6 g/dL. Small bowel resection was indicated, but the attending physician wanted to raise the patient's hemoglobin level to 12 g/dL before surgery. How many units of RBCs would most likely be required to accomplish this?

A. 2
B. 3
C. 6
D. 8

Blood bank/Apply knowledge of fundamental biological characteristics/Blood components/RBCs/2

2. Which of the following is *not* a viable method for removing leukocytes from RBCs?

A. Prestorage filtration
B. Bedside filtration
C. Poststorage filtration
D. Buffy coat removal

Blood bank/Apply knowledge of fundamental biological characteristics/Blood components/RBCs/1

3. Four units of packed RBCs were brought to the nurses' station at 10:20 a.m. Two units were transfused immediately, and 1 unit was transfused at 10:40 a.m. The remaining unit was returned to the blood bank at 11:00 a.m. The units were not refrigerated after leaving the blood bank. What problem(s) is (are) present in this situation?

A. The only problem is with the returned unit; the 30-minute limit has expired and the unit cannot be used
B. The unit should not have been transfused at 10:40 a.m. because the time limit had expired; this unit and the remaining unit should have been returned to the blood bank
C. The returned unit may be held for this patient for 48 hours but cannot be used for another patient
D. No problems; all actions were performed within the allowable time limits

Blood bank/Select course of action/Blood components/RBCs/3

4. A unit of whole blood is collected at 10:00 a.m. and stored at 20°C to 24°C. What is the last hour at which platelet concentrates may be made from this unit?

A. 4:00 p.m.
B. 6:00 p.m.
C. 7:00 p.m.
D. 8:00 p.m.

Blood bank/Apply knowledge of standard operating procedures/Blood components/3

1. **C** One unit of RBCs will raise the hemoglobin level by approximately 1.0 to 1.5 g/dL, and the hematocrit by 3% to 4%. Results vary, depending on the age of the blood and the patient's blood volume and hydration status. Six units will raise the hemoglobin level to at least 12 g/dL.
2. **D** Removal of the buffy coat, which contains both platelets and white blood cells (WBCs), is not an approved method for leukocyte reduction of RBCs. The other methods can be employed to achieve a leukocyte reduction of less than 5×10^6 .
3. **A** There is a 30-minute time limit for a unit of RBCs that is not kept under proper storage conditions (1°C – 6°C).
4. **B** Platelet preparation from whole blood must be done within 8 hours of collection.

5. Which of the following is acceptable according to the AABB standards?

- A. Rejuvenated RBCs may be made within 3 days of outdate and transfused or frozen within 24 hours of rejuvenation
- B. Frozen RBCs must be prepared within 30 minutes of collection and may be used within 10 years
- C. Irradiated RBCs must be treated within 8 hours of collection and transfused within 6 hours
- D. Leukocyte-reduced RBCs must be prepared within 6 hours of collection and transfused within 6 hours of preparation

Blood bank/Apply knowledge of laboratory operations/Blood components/RBCs/2

6. Which of the following is true regarding apheresis platelets?

- A. The minimum platelet count must be 3.0×10^{11} , pH must be 6.0 or greater
- B. The minimum platelet count must be 3.0×10^{10} , pH must be 6.2 or less
- C. The minimum platelet count must be 3.0×10^{11} , pH must be 6.2 or greater
- D. The minimum platelet count must be 5.5×10^{10} , pH must be 6.0 or less

Blood bank/Apply knowledge of laboratory operations/Blood components/Platelets/1

7. What is the component of choice for a patient with chronic granulomatous disease (CGD)?

- A. Fresh frozen plasma (FFP)
- B. Granulocytes
- C. Cryoprecipitate
- D. RBCs

Blood bank/Apply knowledge of laboratory operations/Blood components/3

8. What method can be employed to detect bacteria in random donor platelets?

- A. pH
- B. Glucose
- C. Pan-genera detection (PGD) assay
- D. Gram stain

Blood bank/Apply knowledge of laboratory operations/Blood components/Preparation of components/1

9. All of the following statements regarding FFP are true, except:

- A. FFP must be prepared within 24 hours of collection
- B. After thawing, FFP must be transfused within 24 hours
- C. Storage temperature for FFP with a 1-year shelf life is -18°C or less

D. When thawed, FFP must be stored between 1°C to 6°C

Blood bank/Apply knowledge of standard operating procedures/Blood components/RBCs/1

10. What may be done to RBCs before transfusion to a patient with cold agglutinin disease to reduce the possibility of a transfusion reaction?

A. Irradiate to prevent graft-versus-host-disease (GVHD)

B. Wash with 0.9% saline

C. Warm to 37°C with a blood warmer

D. Transport so that temperature is maintained at 20°C to 24°C

Blood bank/Apply knowledge of standard operating procedures/Hemotherapy/RBCs/2

Answers to Questions 5–10

5. A Rejuvenated RBCs may be prepared within 3 days of the outdate of the unit and washed and transfused or frozen within 24 hours. A unit of RBCs may be frozen within 6 days of collection. An RBC unit can be irradiated any time prior to the expiration date; once irradiated, the unit must be transfused within 28 days of irradiation or the original outdate, whichever comes first. Leukocyte-reduced RBCs should be prepared within 6 hours of collection, but must be given within 24 hours, if prepared using an open system. Leukocyte-reduced RBCs prepared using a closed system may be kept until the original date of expiration.

6. C Single-donor platelets prepared by apheresis must contain a minimum of 3.0×10^{11} platelets, and pH must be 6.2 or higher throughout the shelf life of the product.

7. B Patients with CGD cannot fight bacterial infections because of dysfunctional phagocytic enzymes; granulocyte concentrates are the product of choice for these patients.

8. C The FDA has mandated that pH and glucose can no longer be used as a screening test for platelets. The Verax PGD assay has been FDA approved for both single-donor platelets and random-donor platelets for bacteria screening.

9. A FFP must be prepared within 8 hours after collection or according to FDA-cleared operator manual/inserts.

10. C A patient having cold agglutinins might have a reaction to a cold blood product. The product should be warmed to 37°C before transfusion.

11. A unit of packed RBCs is split using the open system. One of the half units is used. What may be done with the second half unit?

A. Must be issued within 24 hours

B. Must be issued within 48 hours

C. Must be irradiated

D. Must retain the original expiration date

Blood bank/Apply knowledge of laboratory operations/Blood components/RBCs/2

12. What should be done if a noticeable clot is found in an RBC unit?

A. Issue the unit; the blood will be filtered

B. Issue the unit; note the presence of a clot on the release form

C. Filter the unit in the blood bank before issue

D. Do not issue the unit

Blood bank/Select course of action/Hemotherapy/RBCs/2

13. Cryoprecipitate may be used to treat all of the following, except:

A. von Willebrand disease

B. Hypofibrinogenemia

C. Idiopathic thrombocytopenic purpura (ITP)

D. Factor XIII deficiency

Blood bank/Select best course of action/Hemotherapy/Cryo/3

14. SITUATION: A transplant patient may receive only type A or AB platelets. Only type O apheresis platelets are available. What devices may be used to deplete the incompatible plasma and replace with sterile saline?

A. Cytospin/irradiator

B. Water bath/centrifuge

C. Centrifuge/sterile connecting device

D. Cell washer/heat sealer

Blood bank/Apply knowledge of standard operating procedures/Blood components/Platelets/2

15. What component(s) is (are) indicated for patients who have anti-IgA antibodies?

A. Whole blood

B. Packed RBCs

C. Washed or deglycerolized RBCs

D. Granulocytes

Blood bank/Select course of action/Hemotherapy/2

16. What is the expiration date for pooled cryoprecipitate (pooled before freezing)?

A. 12 months from latest date of collection of product in pool

B. 12 months from earliest date of collection of product in pool

C. 4 hours

D. 6 hours

Blood bank/Apply knowledge of standard operating procedures/Blood components/FFP/1

17. All of the following are true regarding washed RBCs, except:

A. RBCs are washed with 1 to 2 L of normal saline

B. Volume is 180 mL

C. Shelf life is extended

D. Leukocytes are removed

Blood bank/Apply knowledge of standard operating procedures/Blood components/Processing/1

Answers to Questions 11–17

11. A The other half unit must be issued within 24 hours, if an open system is used to split the unit.

12. D A unit having a noticeable clot should not be issued for transfusion to a patient. The clot may be an indication of contamination or bacterial growth.

13. C Cryoprecipitate may be used to treat von Willebrand disease, hypofibrinogenemia,

and factor XIII deficiency, but is not indicated in ITP. IVIG is the product of choice for ITP.

- 14. **C** In the event of an ABO-mismatched stem cell transplant, special attention must be paid to the choice of transfused blood products. Type A or AB platelets may be given to a transplant in which the donor is type A and the recipient is type O; once the stem cells engraft, platelets/plasma must be compatible with type A cells. If only type O single-donor platelets are available, the product can be spun down using a centrifuge and plasma can be removed. Then, a sterile connecting device can be used to aseptically transfer sterile isotonic saline to the platelet product, replacing the incompatible plasma.
- 15. **C** Patients with anti-IgA antibodies should not receive components containing plasma. Washed or deglycerolized RBCs can be issued.
- 16. **B** Expiration of pooled cryoprecipitate must take into account that there may be units that makeup the pool with differing collection dates where the earliest date is used to determine the expiration date.
- 17. **C** Washed RBCs renders the system “open” and shortens the expiration time to 24 hours.

18. What is a special condition for the storage of platelets?

- A. Room temperature, 20°C to 24°C
- B. No other components may be stored with platelets
- C. Platelets must be stored upright in separate containers
- D. Platelets require constant agitation at 20°C to 24°C

Blood bank/Apply knowledge of standard operating procedures/Blood components/Processing/1

19. Transfusion of an irradiated product is indicated in all of the following conditions except:

- A. Exchange transfusion
- B. Bone marrow transplantation
- C. Severe combined immunodeficiency syndrome (SCIDS)
- D. Warm autoimmune hemolytic anemia

Blood bank/Select course of action/Hemotherapy/Irradiation/2

20. What percentage of RBCs must be retained in leukocyte-reduced RBCs?

- A. 75%
- B. 80%
- C. 85%
- D. 100%

Blood bank/Apply knowledge of standard operating procedures/Blood components/1

21. Which of the following is true regarding granulocyte concentrates?

- A. The product must contain a maximum of 1.0×10^{10} granulocytes
- B. pH must be 6.0
- C. The product must be crossmatched
- D. The product must be irradiated

Blood bank/Apply knowledge of standard operating procedures/Blood components/2

22. What course of action should be taken if an MLS inadvertently irradiates a unit of RBCs twice?

- A. Issue the unit
- B. Discard the unit
- C. Change the expiration date; then issue the unit
- D. Note on the irradiation sticker that the unit was irradiated twice, and then issue the unit

Blood bank/Apply knowledge of standard operating procedures/Irradiation/2

23. What component(s) may be shipped together with FFP?

- A. Frozen RBCs and cryoprecipitate
- B. Platelets
- C. Packed RBCs and granulocytes
- D. Double RBC units

Blood bank/Apply knowledge of standard operating procedures/Blood components/FFP/1

24. A blood supplier ships 3 units of pooled cryoprecipitate. Each pool consists of 5 units of cryoprecipitate. If one unit is thawed at 5:00 p.m., when must it be dispensed from the blood bank?

- A. Before 9:00 p.m.
- B. Before 11:00 p.m.
- C. Before 12:00 a.m.
- D. Before 5:00 p.m. the next day

Blood bank/Apply knowledge of standard operating procedures/Blood components/FFP/1

Answers to Questions 18–24

18. D Platelets require constant agitation and are stored at temperatures between 20°C and 24°C. Cold-stored platelets (single donor [apheresis] platelets stored at 1–6°C for up to 3 days) may be used for actively bleeding patients, but are not approved for treating thrombocytopenia.

19. D WAIHA would not require irradiation unless the patient had an underlying immunosuppressive disorder.

20. C An RBC unit that has been leukocyte reduced must retain 85% of original RBCs.

21. C Granulocyte concentrates contain a large amount of RBCs (greater than 2 mL) that must be crossmatched with the recipient's serum.

22. B If an MLS mistakenly irradiates a unit of RBCs more than once, the unit must be discarded because of subsequent potassium accumulation. This does not apply to platelets.

23. A FFP requires dry ice for shipment. Frozen RBCs and cryoprecipitate also require dry ice.

24. A A thawed unit of pooled cryoprecipitate must be dispensed within 4 hours unless a sterile connecting device was used, which would extend expiration to 6 hours.

25. How does irradiation prevent transfusion-associated graft-versus-host disease (TA-GVHD)?

- A. Gamma rays and x-rays destroy the lymphocytes' ability to divide
- B. X-rays cause lysis of the lymphocytes
- C. Gamma rays enhance lymphocyte reactivity
- D. Ultraviolet radiation induces apoptosis of lymphocytes

Blood bank/Apply knowledge of standard operating procedures/Blood components/Stem cells/1

26. Which component has the longest expiration date?

- A. Cryoprecipitate
- B. FFP
- C. Frozen RBCs
- D. Platelet concentrates

Blood bank/Apply knowledge of standard operating procedures/Blood bank/Expiration date/1

27. All of the following are advantages of using single-donor platelets as opposed to random donor platelets, *except*:

- A. Less preparation time
- B. Less antigen exposure for patients
- C. May be HLA matched
- D. No pooling is required

Blood bank/Apply principles of special procedures/Blood components/Platelets/1

28. What is the expiration of cryoprecipitate once pooled without the use of a sterile connecting device?

- A. 4 hours
- B. 6 hours
- C. 8 hours
- D. 24 hours

Blood bank/Apply knowledge of standard operating procedures/Blood components/Expiration date/1

29. What is the number of WBCs permitted in a unit of leukocyte-reduced RBCs?

- A. Less than 5×10^{10}
- B. Less than 5×10^6
- C. Less than 8.3×10^5
- D. Less than 8.3×10^6

Blood bank/Apply knowledge of standard operating procedures/Blood components/1

30. SITUATION: A patient with cancer recently developed a severe infection. The patient's hemoglobin level is 8 g/dL as a result of chemotherapy with a drug known to cause bone marrow depression and immunodeficiency. Which blood products are indicated for this patient?

- A. Liquid plasma and cryoprecipitate
- B. Crossmatched platelets and washed RBCs
- C. Factor IX concentrates and FFP
- D. Irradiated RBCs, platelets, and granulocytes

Blood bank/Correlate clinical and laboratory data/Blood and components/3

25. **A** Gamma rays or x-rays have the ability to prohibit a lymphocyte's ability to divide, preventing TA-GVHD.
26. **C** Frozen RBCs may be kept for up to 10 years, or longer if the transfusion service or blood center has a policy for extension beyond 10 years. FFP and cryoprecipitate stored at -18°C or lower expire in 1 year. If FFP is kept at -65°C or lower, the expiration time is 7 years. Platelet concentrates expire from 24 hours to 5 days, depending on the collection system used.
27. **A** Single-donor platelets require more preparation time compared with random-donor platelets because they are prepared by apheresis, which may require 1 to 3 hours, depending on the instrumentation used. Pooling random donor platelets in equivalent amounts may require only a few minutes.
28. **A** When individual cryoprecipitate units are pooled in an open system, the expiration time is 4 hours; if cryoprecipitate is pooled using a sterile connecting device, the expiration time is 6 hours.
29. **B** RBCs that have been leukocyte reduced must have fewer than 5×10^6 WBCs per unit.
30. **D** The patient with cancer may be immunocompromised because of the medication used but needs to receive RBCs for anemia; therefore, irradiated RBCs are indicated. Platelets may be needed to control bleeding, and granulocytes may be indicated for short-term control of severe infection.

4.8

Donors

1. Which of the following individuals is acceptable as a blood donor?

- A. A 29-year-old man who received the hepatitis B vaccine last week
- B. A 21-year-old woman who had her nose pierced last week
- C. A 30-year-old man who lived in Zambia for 3 years and returned last month
- D. A 54-year-old man who tested positive for hepatitis C last year but has no active symptoms of disease

Blood bank/Apply knowledge of standard operating procedures/Donor requirements/2

2. SITUATION: A 53-year-old woman donates blood at her place of employment. She weighs 150 lb and has a hemoglobin level of 13 g/dL. She is currently on warfarin and vitamin B₁₂. Is she an acceptable donor?

- A. Yes
- B. No, she is on warfarin
- C. Yes, for RBCs only
- D. No, her hemoglobin is too low

Blood bank/Apply knowledge of standard operating procedures/Donor requirements/2

3. Which immunization has the longest deferral period?

- A. Hepatitis B immune globulin (HBIG)
- B. Rubella vaccine
- C. Influenza vaccine
- D. Yellow fever vaccine

Blood bank/Apply knowledge of standard operating procedures/Donor requirements/1

4. The following whole blood donors regularly give blood. Which donor may donate on September 10?

- A. A 40-year-old woman who last donated on July 23
- B. A 28-year-old man who had plateletpheresis on August 24
- C. A 52-year-old man who made an autologous donation on September 9
- D. A 23-year-old woman who donated blood for her aunt on August 14

Blood bank/Apply knowledge of standard operating procedures/Donor requirements/2

5. Which of the following precludes acceptance of a plateletpheresis donor?

- A. Platelet count of $75 \times 10^9/L$ in a donor who is a frequent platelet donor
- B. Plasma loss of 800 mL from plasmapheresis 1 week ago
- C. Plateletpheresis performed 4 days ago
- D. Aspirin ingested 7 days ago

Blood bank/Apply knowledge of standard operating procedures/Donor requirements/1

Answers to Questions 1–5

- 1. A** If the donor is symptom free, there is no deferral period for the hepatitis B vaccine.

Persons who have had body piercing are given a 12-month deferral. Persons who lived in an area endemic for malaria or who received antimalarial drugs are deferred for 3 years. A positive test for the hepatitis C virus (HCV) is cause for indefinite deferral.

2. **C** Her age and hemoglobin meet donor criteria. However, because she is currently on warfarin, only RBCs can be prepared from her donation.
3. **A** Deferral for HBIG injection is 12 months. Deferral for rubella vaccine is 4 weeks. The deferral period for influenza and yellow fever vaccines is 2 weeks.
4. **B** A plateletpheresis donor must wait at least 48 hours between donations. The waiting period following an autologous donation is at least 3 days. An 8-week interval must pass between all other types of donations.
5. **A** To be eligible for plateletpheresis, the platelet count should be greater than $150 \times 10^9/L$ for a frequent platelet donor. Plasma loss exceeding 1,000 mL would be cause for rejection, but not 800 mL. A donor may donate 24 times a year, but not as frequent as once every 2 days in a 7-day period. A donor cannot ingest aspirin within 36 hours of platelet donation.

6. Which of the following donors could be accepted for whole blood donation?

- A. A construction worker who was incarcerated for opiate abuse
- B. A triathlete with a pulse of 45
- C. A man who is currently taking finasteride (Propecia)
- D. A woman in her 14th week of pregnancy

Blood bank/Apply knowledge of standard operating procedures/Donor requirements/2

7. Which physical examination result is cause for rejecting a whole blood donor?

- A. Weight of 105 lb
- B. Pulse of 75
- C. Temperature of 37.4°C (99.3°F)
- D. Diastolic pressure of 110 mm Hg

Blood bank/Apply knowledge of standard operating procedures/Donor requirements/1

8. Which situation is *not* a cause for indefinite deferral of a donor?

- A. Male donor whose father has been diagnosed with Creutzfeldt-Jakob disease (CJD)
- B. Donation of a unit of blood that transmitted hepatitis B virus (HBV) to a recipient
- C. Donor tested reactive for HBV by nucleic acid testing (NAT)
- D. Accidental needlestick 1 year ago; negative for infectious disease

Blood bank/Apply knowledge of standard operating procedures/Donor requirements/1

9. A whole blood donor currently on clopidogrel (Plavix) is precluded from donating which product?

- A. Platelets
- B. RBCs
- C. FFP
- D. Cryoprecipitate

Blood bank/Select course of action/Donor processing/Unacceptable donors/1

10. How much anticoagulant would have to be removed from the collection bag given a donor who weighs 90 lb?

- A. 12 mL

- B. 15 mL
- C. 20 mL
- D. 23 mL

Blood bank/Apply knowledge of standard operating procedures/Donor collection/2

11. A woman begins to breathe rapidly while donating blood. Choose the correct course of action.

- A. Continue the donation; rapid breathing is not a reason to discontinue a donation
- B. Withdraw the needle, raise her feet, and administer ammonia
- C. Discontinue the donation, and provide a paper bag for donor to breathe slowly into
- D. Tell her to sit upright, and apply a cold compress to her forehead

Blood bank/Select course of action/Donor processing/Donor adverse reactions/3

Answers to Questions 6–11

- 6. **B** Athletes may have a pulse below 50 and may still be acceptable as blood donors. Drug addiction is cause for permanent deferral, as is a major illness. The deferral period after treatment for syphilis or gonorrhea is 12 months.
- 7. **D** Diastolic pressure must not be higher than 100 mm Hg. Donors weighing less than 110 lb may donate up to 12% of their blood volume (volume = weight in kg/50 × 450 mL). Oral temperature must not be greater than 37.4°C (99.5°F). BP limits for donation are 180 mm Hg for systolic and 100 mm Hg for diastolic pressure. The limit for hemoglobin is 12.5 g/dL and for hematocrit 38%.
- 8. **D** An accidental needlestick would not be a cause for indefinite deferral of a donor. The deferral period is 1 year.
- 9. **A** Clopidogrel (Plavix) renders platelets nonfunctional, and therefore potential donors on this medication cannot donate platelets.
- 10. **A** To determine the amount of anticoagulant to remove when the donor is less than 110 lb, divide weight by 110 lb, and multiply by 450 mL; divide that number by 100, and multiply by 14 (this gives the anticoagulant volume needed); then subtract this from 63 mL, which is the standard volume of anticoagulant in a 450 mL bag. The result is the amount of anticoagulant to remove.
- 11. **C** This woman is hyperventilating; therefore, the donation should be discontinued. A paper bag should be provided to the donor to breathe into to increase the carbon dioxide in the donor's air.

12. A donor bag is half filled during donation when the blood flow stops. Select the correct course of action.

- A. Closely observe the bag for at least 3 minutes; if blood flow does not resume, withdraw the needle
- B. Remove the needle immediately, and discontinue the donation
- C. Check and reposition the needle, if necessary; if blood flow does not resume, withdraw the needle
- D. Withdraw the needle, and perform a second venipuncture in the other arm

Blood bank/Select course of action/Collection/3

13. Who is the best candidate for a predeposit autologous donation?

- A. A 45-year-old man who is having elective surgery in 2 weeks; he has alloanti-k
- B. A 23-year-old female patient with leukemia and a hemoglobin level of 10 g/dL
- C. A 12-year-old boy who has hemophilia
- D. A 53-year-old woman who has septicemia

Blood bank/Select course of action/Donor processing/Autologous donation/2

14. Can an autologous donor donate blood on Monday if he or she is having surgery on Friday?

- A. Yes, he or she can donate up to 72 hours before surgery
- B. No, he or she cannot donate within 7 days of surgery
- C. Yes, he or she can donate, but only a half a unit
- D. No, he or she cannot donate within 5 days of surgery

Blood bank/Apply knowledge of standard operating procedures/Autologous donation/2

15. Which of the following is an acceptable time in which a unit of whole blood is collected?

- A. 33 minutes
- B. 25 minutes
- C. 20 minutes
- D. 13 minutes

Blood bank/Apply knowledge of standard operating procedures/Collection/1

16. Which of the following is true regarding acute normovolemic hemodilution?

- A. One or more units of blood are withdrawn from the patient and replaced with FFP
- B. Units removed may be stored in the operating room at room temperature for 8 hours
- C. Units removed may be stored in the operating room at room temperature for 24 hours
- D. Unused units can be added to the general donor blood inventory

Blood bank/Apply knowledge of standard operating procedures/Autologous donation/2

17. All of the following apply to a double RBC unit apheresis collection except:

- A. The hematocrit must be at least 38%
- B. The weight for a female is at least 150 lb
- C. The height for a male is at least 5 ft 1 in
- D. The deferral period following collection is 16 weeks

Blood bank/Apply knowledge of standard operating procedures/Apheresis/1

Answers to Questions 12–17

- 12. **C** If blood flow has stopped, check the needle first. If blood flow does not resume after repositioning, then withdraw the needle and discontinue the donation. Do not perform a second venipuncture on the donor.
- 13. **A** The 45-year-old man with alloanti-k is the best candidate for predeposit autologous donation because compatible blood will be hard to find if he needs blood after surgery. The other candidates may not be good choices for donation because the process may prove harmful to them.
- 14. **A** An autologous donor can donate up to 72 hours before expected surgery.
- 15. **D** A unit of whole blood should be collected within 15 minutes.
- 16. **B** In acute normovolemic hemodilution, one or more units of blood are removed from

the donor and replaced with crystalloid or colloid. Blood may be stored at room temperature for up to 8 hours or at 1°C to 6°C for up to 24 hours. Bleeding during surgery results in less RBC loss after hemodilution, and the autologous RBCs are infused after bleeding stops. Such units are for autologous transfusion only.

17. A The minimum hematocrit for a double RBC donation is 40%.

18. An autologous unit of whole blood was collected on a 33-year-old woman in preparation for a knee replacement procedure in 3 weeks. The whole blood unit had her hyphenated last name, first name, and last four digits of her social security number for identification. The laboratory computer system, however, only had her married name and first name, medical record number, and social security number. What should be done with this blood product?

- A.** Discard the unit
- B.** Make the unit available for transfusion
- C.** Confirm the name with the donor, have admissions make the correction in the computer system, and then make the unit available for transfusion
- D.** Ensure that social security numbers match, confirm the name with the donor and have admissions make the correction in the computer system with the medical director's approval, and then make the unit available for transfusion

Blood bank/Standard operating procedures/Autologous donation/3

19. What is the youngest age a person can make allogeneic whole blood donation?

- A.** 14 years
- B.** 15 years
- C.** 16 years
- D.** 17 years

Blood bank/Apply knowledge of standard operating procedures/Donation/1

20. Which of the following donors are acceptable for whole blood donation?

- A.** A man who had sex with another man 2 weeks ago
- B.** A male construction worker who tested positive for hepatitis B surface antigen (HBsAg)
- C.** A man who had sex with another man 2 years ago
- D.** A female who had sex with a male who had sex with another male (MSM) 1 month ago

Blood bank/Apply knowledge of standard operating procedures/Donors/1

Answers to Questions 18–20

18. D This is a common scenario with women who have recently married, and have not changed their license or other form of identification given to the collection facility. Checking that other demographic information matches is sufficient, if approved by the medical director, because an autologous unit is very difficult to replace in time for surgery.

19. C In most states, the youngest age a person can donate is 16 years with parental permission.

20. C Men who have sex with men can donate blood if they have not had sex with another man in the past year.

Hemolytic Disease of the Fetus and Newborn

1. **All of the following are reasons for a positive DAT on cord blood cells of a newborn except:**
 - A. High concentrations of Wharton jelly on cord blood cells
 - B. Immune anti-A from an O mother on the cells of an A baby
 - C. Immune anti-D from an Rh negative mother on the cells of an Rh-positive baby
 - D. Immune anti-K from a K-negative mother on the cells of a K-negative baby

Blood bank/Correlate clinical and laboratory data/Hemolytic disease of the fetus and newborn/DAT/2
2. **A fetal screen yielded negative results on a mother who is O negative and infant who is O positive. What course of action should be taken?**
 - A. Perform a Kleihauer-Betke test
 - B. Issue one full dose of RhIg
 - C. Perform a DAT on the infant
 - D. Perform an antibody screen on the mother

Blood bank/Select course of action/Hemolytic disease of the fetus and newborn/Rosette test/3
3. **What should be done when a woman who is 24 weeks pregnant has a positive antibody screen?**
 - A. Perform an antibody identification panel; titer, if necessary
 - B. No need to do anything until 30 weeks' gestation
 - C. Administer RhIg
 - D. Adsorb the antibody onto antigen-positive cells

Blood bank/Apply knowledge of standard operating procedures/Hemolytic disease of the fetus and newborn/Antibody testing/2
4. **All of the following are interventions for fetal distress caused by maternal antibodies attacking fetal cells except:**
 - A. Intrauterine transfusion
 - B. Plasmapheresis on the mother
 - C. Transfusion of antigen-positive cells to the mother
 - D. Middle cerebral artery peak systolic velocity (MCA-PSV)

Blood bank/Apply knowledge of standard operating procedures/Hemolytic disease of the fetus and newborn/Clinical interventions/2
5. **Cord blood cells are washed six times with saline, and the DAT result and negative control are still positive. What should be done next?**
 - A. Obtain a heelstick sample
 - B. Record the DAT result as positive
 - C. Obtain another cord blood sample
 - D. Perform elution on the cord blood cells

Blood bank/Select course of action/Hemolytic disease of the fetus and newborn/DAT/3

Answers to Questions 1–5

1. **D** Immune anti-K from the mother would not coat the baby's RBCs if they did not contain the K antigen; therefore, the DAT result would be negative.
2. **B** If the fetal screen or rosette test has a negative result, indicating the fetal maternal blood is negligible in a possible RhIg candidate, standard practice is to issue one dose of RhIg.
3. **A** The identification of the antibody is very important at this stage of the pregnancy. If the antibody is determined to be clinically significant, then a titer may determine the strength of the antibody and the need for clinical intervention.
4. **C** Transfusion of antigen-positive cells to the mother who already has an antibody might cause a transfusion reaction and/or evoke an even stronger antibody response, possibly causing more harm to the fetus.
5. **A** If the cord blood cells contain excessive Wharton jelly, then further washing or obtaining another cord blood sample will not solve the problem. A heelstick sample will not contain Wharton jelly and should give a valid DAT result.

6. What can be done if HDFN is caused by maternal anti-K?

- A. Give Kell immune globulin
- B. Monitor the mother's antibody level
- C. Prevent formation of K-positive cells in the fetus
- D. Not a problem; anti-K is not known to cause HDFN

Blood bank/Apply principles of special procedures/Hemolytic disease of the fetus and newborn/Antibody formation/2

7. Should an O-negative mother receive RhIg if a positive DAT on the newborn is caused by immune anti-A?

- A. No, the mother is not a candidate for RhIg because of the positive DAT result
- B. Yes, if the baby's type is Rh negative
- C. Yes, if the baby's type is Rh positive
- D. No, the baby's problem is unrelated to Rh blood group antibodies

Blood bank/Correlate clinical and laboratory data/Hemolytic disease of the fetus and newborn/RhIg/3

8. Should an A-negative woman who has just had a miscarriage receive RhIg?

- A. Yes, but only if she does not have evidence of active anti-D
- B. No, the type of the baby is unknown
- C. Yes, but only a minidose regardless of trimester
- D. No, RhIg is given to women at full-term pregnancies only

Blood bank/Apply knowledge of standard operating procedures/Hemotherapy/RhIg/3

9. SITUATION: The automated blood bank analyzer reports a type of O negative on a woman who is 6 weeks pregnant with vaginal bleeding. The woman tells the emergency department physician she is O positive and presents a blood donor card. The MLS performs a test for weak D and observes a 1+ reaction in the AHG phase. The Kleihauer-Betke test result is negative. Is this woman a candidate for RhIg?

A. Molecular testing is indicated to ascertain the type of weak D

B. Yes, she is Rh positive

C. No, there is no evidence of a fetal bleed

D. Yes, based on the automated typing results

Blood bank/Correlate clinical and laboratory results/Hemolytic disease of the fetus and newborn/RhIg/3

10. Which of the following patients would be a candidate for RhIg?

A. B-positive mother; B-negative baby; first pregnancy; no anti-D in mother

B. O-negative mother; A-positive baby; second pregnancy; no anti-D in mother

C. A-negative mother; O-negative baby; fourth pregnancy; anti-D in mother

D. AB-negative mother; B-positive baby; second pregnancy; anti-D in mother

Blood bank/Correlate clinical and laboratory data/Hemolytic disease of the fetus and newborn/RhIg/2

Answers to Questions 6–10

6. **B** Anti-D is the only antibody for which prevention of HDFN is possible. If a pregnant woman develops anti-K, she will be monitored to determine if the antibody level and signs of fetal distress necessitate clinical intervention.

7. **C** RhIg is immune anti-D and is given to Rh-negative mothers who give birth to Rh-positive babies and who do not have anti-D already formed from previous pregnancies or transfusion.

8. **A** When the fetus is Rh positive or the Rh status of the fetus is unknown, termination of a pregnancy from any cause presents a situation in which an Rh-negative patient should receive RhIg. A minidose is used if the pregnancy is terminated in the first trimester.

9. **A** The negative Kleihauer-Betke test result confirms that the positive reaction of the woman's RBCs with anti-D at the IAT phase is not the result of fetal–maternal bleeding. The woman is weak D positive and, therefore, requires molecular testing to discern if she is a candidate for RhIg. Weak D types 1, 2, 3, and 4 do not make anti-D and do not require RhIg prophylaxis.

10. **B** An O-negative mother who gives birth to an A-positive baby and has no anti-D formed from a previous pregnancy would be a candidate for RhIg. A mother who already has active anti-D or a mother who gives birth to an Rh-negative baby is not a candidate for RhIg. Anti-D formation via active immunization typically has a titer greater than 4, compared with passive administration of anti-D, which has a titer less than 4.

11. The Kleihauer-Betke acid elution test identifies 40 fetal cells in 2,000 maternal RBCs. How many full doses of RhIg are indicated?

A. 1

B. 2

C. 3

D. 4

Blood bank/Calculate/Hemolytic disease of the fetus and newborn/RhIg/2

12. Kernicterus is caused by the effects of:

- A. Anemia
- B. Unconjugated bilirubin
- C. Antibody specificity
- D. Antibody titer

Blood bank/Apply knowledge of biological principles/Hemolytic disease of the fetus and newborn/1

13. Anti-E is detected in the serum of a woman in the first trimester of pregnancy. The first titer for anti-E is 32. Two weeks later, the antibody titer is 64 and then 128 after another 2 weeks. Clinically, there are beginning signs of fetal distress. What may be done?

- A. Induce labor for early delivery
- B. Perform plasmapheresis to remove anti-E from the mother
- C. Administer RhIg to the mother
- D. Perform an intrauterine transfusion using E-negative cells

Blood bank/Correlate clinical and laboratory data/Hemolytic disease of the fetus and newborn/3

14. What testing is done for exchange transfusion when the mother's serum contains an alloantibody?

- A. Crossmatching and antibody screen
- B. ABO, Rh, antibody screen, and crossmatching
- C. ABO, Rh, antibody screen
- D. ABO and Rh only

Blood bank/Apply knowledge of standard operating procedures/Hemolytic disease of the fetus and newborn/Hemotherapy/1

15. Which blood type may be transfused to an AB-positive baby who has HDFN caused by anti-D?

- A. AB negative, CMV negative, Hgb S negative; irradiated or O negative, CMV negative, Hgb S negative
- B. AB positive, CMV negative; irradiated or O positive, CMV negative
- C. AB negative only
- D. O negative only

Blood bank/Select course of action/Hemolytic disease of the fetus and newborn/Hemotherapy/2

16. All of the following are routinely performed on a cord blood sample except:

- A. Forward ABO typing
- B. Antibody screen
- C. Rh typing
- D. DAT

Blood bank/Apply knowledge of laboratory operations/Hemolytic disease of the fetus and newborn/Cord blood/1

Answers to Questions 11–16

11. D To calculate the number of vials of RhIg to infuse, divide 40 by 2,000, and multiply

by 5,000. This gives the estimated total blood volume of the mother in milliliters. Divide this number by 30 to obtain the number of doses. When the number to the right of the decimal point is less than 5, round down, and add one dose of RhIg. Conversely, when the number to the right of the decimal point is 5 or greater, round up, and add one dose of RhIg. In this example, the number of doses is 3.3. Rounding down and adding one vial gives an answer of four vials.

- 12. **B** Kernicterus occurs because of high levels of unconjugated bilirubin. High levels of this pigment cross into the central nervous system, causing brain damage to the infant.
- 13. **B** Plasmapheresis removes excess anti-E from the mother and provides a temporary solution to the problem until the fetus is mature enough to be delivered. The procedure may need to be performed several times, depending on how quickly and how high the levels of anti-E rise. Administration of RhIg would not contribute to solving this problem caused by anti-E. Intrauterine transfusion would not be performed before week 20 and would be considered only if there is evidence of severe hemolytic disease.
- 14. **B** ABO (forward) and Rh are required. An antibody screen using either the neonatal serum or maternal serum is required. Crossmatching is necessary as long as maternal antibody persists in the infant's blood.
- 15. **A** Either AB-negative or O-negative RBCs may be given to an AB-positive baby because both types are ABO compatible and lack the D antigen.
- 16. **B** An antibody screen is not performed routinely on a cord blood sample because a baby does not make antibodies until about 6 months of age. Any antibodies detected in a cord blood sample come from the mother.

17. Why do Rh-negative women tend to have a positive antibody screen compared with Rh-positive women of childbearing age?

- A. They have formed active anti-D
- B. They have received RhIg
- C. They have formed anti-K
- D. They have a higher rate of transfusion

Blood bank/Apply knowledge of biological principles/Hemolytic disease of the fetus and newborn/2

18. SITUATION: An O-negative mother gave birth to a B-positive infant. The mother had no history of antibodies or transfusion. This was her first child. The baby was mildly jaundiced, and the DAT result was weakly positive with polyspecific antisera. What could have caused the positive DAT result?

- A. Anti-D from the mother coating the infant RBCs
- B. An alloantibody, such as anti-K, coating the infant RBCs
- C. Maternal anti-B coating the infant RBCs
- D. Maternal anti-A, B coating the infant RBCs

Blood bank/Correlate clinical and laboratory data/Hemolytic disease of the fetus and newborn/3

19. SITUATION: RhIg is requested on a 28-year-old woman with suspected abortion. When the nurse arrives in the blood bank to pick up the RhIg, she asks the MLS if it is a minidose. The MLS replies that it is a full dose, not a minidose. The nurse then

requests to take 50 µg from the 300 µg syringe to satisfy the physician's orders. What course of action should the MLS take?

- A.** Let the nurse take the syringe of RhIg, so that she may withdraw 50 µg
- B.** Call a supervisor or pathologist
- C.** Instruct the nurse that the blood bank does not stock minidoses of RhIg and manipulating the full dose will compromise the purity of the product
- D.** Instruct the nurse that the blood bank does not stock minidoses of RhIg, and relay this information to the patient's physician

Blood bank/Select course of action/Hemolytic disease of the fetus and newborn/RhIg/3

Answers to Questions 17–19

- 17. B** The most common reason an Rh-negative woman has a positive antibody screen is because of previously receiving RhIg or passive anti-D.
- 18. D** Anti-A,B is an IgG antibody and can cross the placenta and attach to infant cells. It is known as a single entity as opposed to separate antibodies. Anti-D would not be the cause because this is the first pregnancy. Anti-K is not the cause because there is no history of alloantibodies or past transfusions.
- 19. D** Blood banks operate by strict standard operating procedures. These include decisions about which products are supplied from the blood bank. Although B may also be a solution, D is the best answer because the patient's physician can communicate with the pathologist once he or she receives this information from the nurse.

4.10

Serological Testing of Blood Products

1. **What protocol is followed when screening whole blood donors for HIV-1 RNA?**
 - A. Pools of 10 are tested; if the pool is nonreactive, donors are accepted
 - B. Pools of 20 are tested; if the pool is reactive, samples are tested individually
 - C. Pools of up to 16 donors are tested; if pool is reactive, individual samples are screened
 - D. All donors are screened individually; if samples are reactive, blood is discarded

Blood bank/Standard operating procedures/Processing/3

2. **Currently, nucleic acid testing (NAT) testing is performed to detect which viruses?**
 - A. HIV and Human T-cell lymphotropic virus (HTLV-1)
 - B. HTLV I/II
 - C. HIV, HCV, HBV, Zika, and West Nile virus (WNV)
 - D. HIV, HBV, and WNV

Blood bank/Apply knowledge of standard operating procedures/Processing/1

3. **John comes in to donate a unit of whole blood at the collection center of the community blood supplier. The enzyme immunoassay assay (EIA) screen for anti-HIV-1,2 is nonreactive; however, the NAT HIV is reactive. After 8 weeks John is tested again and found to be NAT nonreactive and EIA anti-HIV-1,2 nonreactive:**
 - A. This renders him eligible for donation
 - B. Donation is deferred for 6 months
 - C. Status is dependent on further confirmatory testing
 - D. Donation is deferred for 12 months

Blood bank/Select course of action/Processing/3

4. **What marker is the first to appear in HBV infection?**
 - A. Hepatitis B core antibody (anti-HBc) IgM
 - B. HbsAg
 - C. Hepatitis B surface antibody (anti-HBs)
 - D. Anti-HBc IgG

Blood bank/Apply knowledge of biological principles/Processing/1

5. **What marker indicates immunity to hepatitis B infection?**
 - A. Anti-HBc IgM
 - B. HBsAg
 - C. Anti-HBs
 - D. Anti-HBc IgG

Blood bank/Apply knowledge of standard operating procedures/Processing/1

Answers to Questions 1–5

1. **C** Pools of up to 16 donors are tested by nucleic acid amplification technology. If the pool is reactive, samples from each individual donor are tested. Pooled donors may

also be screened for Zika virus using nucleic acid amplification.

2. **C** According to the AABB standards, NAT testing is required for viruses HIV-1, HCV, HBV, Zika, and WNV.
3. **A** According to the FDA guidelines, donors who retest as NAT HIV nonreactive and negative for anti-HIV-1,2 are eligible to donate in their re-entry program.
4. **B** The first viral marker of hepatitis B to appear in the serum once exposed is the HBSAg, which appears in as few as 5 days (5–28 days) after exposure.
5. **C** Anti-HBs is indicative of immunity or vaccination to hepatitis B. Anti-HBc IgM occurs in the early stage of infection; anti-HBc IgG follows and may persist for years following infection. HBsAg is a marker of HBV infection, not immunity.

6. **A patient with multiple myeloma is placed on daratumumab (Darzalex). What tests are affected by this drug, and what are typical recommendations for transfusion?**

- A. Antibody screen/least incompatible RBCs
- B. ABO/washed RBCs
- C. Rh/leukocyte-poor RBCs
- D. All of the above

Blood bank/Select best course of action/Processing/3

7. **A unit tests positive for syphilis with use of the rapid plasma reagin (RPR) test. The *Treponema pallidum* particle agglutination (TP-PA) test on the same unit is negative. What is the disposition of the unit?**

- A. The unit may be used to prepare components
- B. The donor must be contacted and questioned further; if the RPR test result is most likely a false positive, then the unit may be used
- C. The unit must be discarded
- D. Cellular components may be prepared but must be irradiated before issue

Blood bank/Apply knowledge of standard operating procedures/Processing/2

8. **SITUATION: John Smith donated a unit of whole blood in May. RBCs made from whole blood were transfused to a recipient at a community hospital in June, with no apparent complications. The blood supplier notified the medical director of the hospital that the donor had reported engaging in high-risk behavior with another male in April, although viral tests remain negative and the donor is healthy. What course of action should be taken?**

- A. No action should be taken
- B. The recipient's physician should be notified
- C. The recipient's physician and the recipient should be notified
- D. The recipient should be notified

Blood bank/Apply knowledge of biological principles/Market withdrawal/3

9. **All of the following are required tests on donor blood, except:**

- A. HBsAg
- B. Anti-CMV
- C. HIV-1
- D. Anti-HTLV I/II

Blood bank/Apply knowledge of standard operating procedures/Processing/1

10. Which of the following bands would constitute a positive Western Blot result for HIV?

- A. p24, gp41, p17
- B. p55, gp120, p51
- C. gp160, p31, p56
- D. p24, p30, p55

Blood bank/Apply knowledge of standard operating procedures/Processing/1

Answers to Questions 6–10

- 6. **A** Daratumumab (Darzalex) is a monoclonal antibody directed against CD38, which shows enhanced expression on multiple myeloma tumor cells. Because CD38 is also expressed on RBCs, all screening cells will be positive. The serological workup appears as a warm autoantibody that cannot be resolved with adsorptions and often necessitates the transfusion of least incompatible RBCs.
- 7. **A** This is a case of a false-positive screening test result (RPR). The confirmatory test for treponemal antibodies was negative. The donor unit is acceptable and may be used to prepare blood components.
- 8. **B** The recipient's physician should be notified by the medical director to ascertain the current health status of the recipient, if known, and determine what treatment, if any, the recipient should receive.
- 9. **B** Testing of donor blood for antibodies to CMV is not required. However, testing may be done on units intended for transfusion to low-birth-weight infants born to seronegative mothers or units used for intrauterine transfusion; units intended for immunocompromised patients who are seronegative; prospective transplant recipients who are seronegative; or transplant recipients who have received a seronegative organ. Leukocyte-reduced RBCs carry a reduced risk of transmitting CMV and are recommended for such patients when CMV testing has not been performed on donor units. The prevalence of anti-CMV in the population ranges from 40% to 90%.
- 10. **A** According to current FDA and CDC criteria, a sample is defined as anti-HIV positive if at least two of the following bands are present on Western blot: p24, gp41, and/or GP120/160.

4.11

Immunohematology Problem-Solving

1. Is there a discrepancy between the following blood typing and secretor study results?

Blood typing results:	Anti-A	Anti-B	A ₁ Cells	B Cells
	4+	0	0	4+
Secretor results:	Anti-A + saliva + A ₁ cells = 0			
	Anti-B + saliva + B cells = 4+			
	Anti-H + saliva + O cells = 0			

- A. No problem, the sample is from a group A secretor
- B. Blood types as A and saliva types as B
- C. Blood types as A, but the secretor study is inconclusive
- D. No problem, the sample is from a group A nonsecretor

Blood bank/Evaluate laboratory data to make identifications/Saliva neutralization/3

2. What is the best course of action given the following test result? (Assume the patient has not been transfused recently.)

Anti-A	Anti-B	A ₁ cells	B cells
Mixed field	0	1+	4+

- A. Nothing, typing is normal
- B. Type patient cells with anti-A₁ lectin and type serum with A₂ cells
- C. Retype patient cells; type with anti-H and anti-A,B; use screen cells or A₂ cells on patient serum; run patient autocontrol
- D. Wash patient cells four times with saline; then repeat the forward type

Blood bank/Apply principles of special procedures/RBCs/ABO discrepancy/3

3. The following results were obtained on a 41-year-old female:

Anti-A	Anti-B	A ₁ cells	B cells	O cells
4+	0	3+	4+	3+

Due to the discrepant reverse grouping, a panel was performed on patient serum revealing the presence of anti-M. How can the reverse grouping be resolved?

- A. Repeat the reverse grouping with a 10-minute incubation at room temperature
- B. Repeat the reverse grouping using A₁ cells that are negative for M antigen
- C. Repeat the reverse grouping using A₁ cells that are positive for M antigen
- D. No further work is necessary

Blood bank/Evaluate laboratory data to recognize problems/ABO discrepancy/3

1. **A** The blood typing result demonstrates A antigen on the RBCs and anti-B in serum. The secretor result reveals A antigen in the saliva. A antigen neutralized anti-A, preventing agglutination when A₁ cells were added. Each blood type (except a Bombay phenotype) contains some H antigen; therefore, H antigen in saliva would be bound by anti-H reagent. No agglutination would occur when O cells are added.
 2. **C** The mixed-field reaction with anti-A suggests a subgroup of A, most likely A₃. The reverse grouping shows weak agglutination with A₁ cells, indicating anti-A₁. A positive reaction with anti-A,B would help to differentiate an A subgroup from group O. If A₂ cells are not agglutinated by patient serum, the result would indicate the presence of anti-A₁. If the patient's serum agglutinates A₂ cells, then an alloantibody or autoantibody should be considered.
 3. **B** The scenario showed an antibody in the patient serum directed toward M antigen, and the M antigen happened to be on the A₁ cells in reverse grouping. To solve this problem, find A₁ cells negative for M antigen or enzyme treat the A₁ cells to resolve the ABO discrepancy.
4. **A 59-year-old male came to the emergency department of a community hospital complaining of dizziness and fatigue. History included no transfusions and a positive rheumatoid factor 1 year ago. His complete blood count (CBC) confirmed anemia. A sample was sent to the blood bank for typing and crossmatching. Upon receipt of the sample in the blood bank, the MLS noticed the ethylenediaminetetraacetic acid (EDTA) sample appeared very viscous. Fearing that the sample would clog the automated instrument, testing was performed by using the tube method. The initial results revealed the following:**

Anti-A	Anti-B	Anti-D	Rh Control	A ₁ cells	B cells
0	0	4+	2+	4+	4+

The patient's RBCs were washed eight times with saline, and testing was repeated, giving the following results:

Anti-A	Anti-B	Anti-D	Rh Control	A ₁ cells	B cells
0	0	4+	0	4+	4+

The antibody screen was negative at the IS, 37°C, and AHG phases; the check cells were positive. Crossmatch testing using two O-positive donor units revealed 1+ at IS, and negative results at the 37°C and AHG phases. The check cells were positive. In light of the crossmatching results, what is the next course of action?

- A. Use other donor cells for crossmatching
- B. Perform a saline replacement for crossmatching
- C. Run the crossmatching test using the Gel system
- D. Result the crossmatch as incompatible

Blood bank/Correlate clinical and laboratory data/Rh discrepancy/3

5. **The following results were obtained on a 51-year-old male with hepatitis C:**

Anti-A	Anti-B	Anti-D	A ₁ cells	B cells
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What should be done next?

- Retype the patient's sample to confirm group AB positive
- Repeat the Rh typing
- Run a saline control in forward grouping
- Report the patient as group AB, Rh positive

Blood bank/Apply knowledge of routine laboratory procedures/ABO/2

Anti-D	Anti-C	Anti-E	Anti-c	Anti-e
4+	2+	0	0	3+

Blood bank/Apply knowledge of fundamental biological characteristics/Rh typing/3

- A.** She has developed an antibody to fetal RBCs
- B.** She probably does not have antibodies because this is her first pregnancy, and she has not been transfused; check for technical error
- C.** She received an antenatal dose of RhIg
- D.** Impossible to determine without further testing

4. **B** The history of the patient correlates with abnormal plasma proteins causing a positive result with the Rh control. Perform a saline replacement technique to rectify the incompatible crossmatches at IS.
5. **C** In the case of an AB-positive person, a saline control must be run in forward grouping to obtain a negative reaction; this will ensure agglutination is specific in the other reactions.
6. **C** The most likely genotype is R^1R^1 . The possibilities are DCe/DCe or DCe/dCe , which translates to R^1R^1 or R^1h' . The former is more common.
7. **C** Because the patient has never been transfused or pregnant previously, she probably has not formed any atypical antibodies. Because she is Rh negative she would have received a dose of RhIg at 28 weeks (antenatal dose) if her prenatal antibody screen had been negative. Although technical error cannot be ruled out, it is far less likely than RhIg administration to be the cause.

8. A patient's serum contains a mixture of antibodies. One of the antibodies is identified as anti-D. Anti-Jk^a or anti-Fy^a and possibly another antibody are present. What technique(s) may be helpful to identify the other antibody(s)?
- A. Enzyme panel; select cell panel
 - B. Thiol reagents
 - C. Lowering the pH and increasing the incubation time
 - D. Using albumin as an enhancement medium in combination with selective adsorption
- Blood bank/Apply principles of special procedures/Antibody ID/3*
9. An anti-M reacts strongly through all phases of testing. Which of the following techniques would *not* contribute to removing this reactivity so that more clinically significant antibodies may be revealed?
- A. Acidifying the serum
 - B. Prewarmed technique
 - C. Adsorption with homozygous cells
 - D. Testing with enzyme-treated RBCs
- Blood bank/Apply principles of special procedures/Antibody ID/3*
10. The reactivity of an unknown antibody could be anti-Jk^a, but the antibody identification panel does not fit this pattern conclusively. Which of the following would *not* be effective in determining if the specificity is anti-Jk^a?
- A. Testing with enzyme-treated cells
 - B. Select panel of homozygous cells
 - C. Testing with 2-aminoethylisothiuronium bromide (AET)-treated cells
 - D. Increased incubation time
- Blood bank/Apply principles of special procedures/Antibody ID/3*
11. A cold-reacting antibody is found in the serum of a recently transfused patient and is suspected to be anti-I. The antibody identification panel shows reactions with all cells at room temperature, including the autocontrol. The reaction strength varies from 2+ to 4+. What procedure would help to distinguish this antibody from other cold-reacting antibodies?
- A. Autoadsorption technique
 - B. Neutralization using saliva
 - C. Autocontrol using ZZAP reagent-treated cells
 - D. Reaction with cord blood cells
- Blood bank/Apply principles of special procedures/Antibody ID/3*

Answers to Questions 8–11

8. **A** An enzyme panel would help to distinguish between anti-Jk^a (reaction enhanced) and anti-Fy^a (destroyed). Anti-D, however, would also be enhanced and may mask reactions that may distinguish another antibody. A select panel of cells negative for D may help to reveal an additional antibody or antibodies.
9. **A** Lowering the pH will actually enhance reactivity of anti-M. Prewarming (anti-M is a cold-reacting antibody), cold adsorption with homozygous M cells, and testing the serum with enzyme-treated RBCs (destroys M antigens) are all techniques to remove reactivity of anti-M.

- 10. C** AET denatures Kell antigens and has no effect on Kidd antibodies. Because the detection of Kidd antibodies is subject to dosage effect, selection of cells homozygous for the Jk^a antigen (and longer incubation) would help to detect the presence of the corresponding antibody. Enzyme-treated RBCs would also react more strongly in the presence of Kidd antibodies.
- 11. D** Because RBCs contain variable amounts of I antigen, reactions with anti-I often vary in agglutination strength. However, because this patient was recently transfused, the variation in reaction strength may be the result of an antibody mixture. Although autoadsorption would remove anti-I, this procedure does not confirm the antibody specificity and can result in removal of other antibodies, as well. Cord blood cells express primarily i antigen with very little I antigen. Anti-I would react weakly or negatively with cord RBCs. ZZAP removes IgG antibodies from RBCs. Because anti-I is IgM, the use of ZZAP would not be of value.

- 12. An antibody identification panel reveals the presence of anti-Le^b and a possible second specificity. Saliva from which person would best neutralize the Le^b antibody?**

	Genes		
	Lewis	ABO	Secretor
A.	Le	H	sese
B.	Le	hh	Se
C.	Le	H	Se
D.	lele	hh	sese

Blood bank/Apply principles of special procedures/Antibody ID/3

- 13. The automated blood bank analyzer does not detect weak forms of D antigen. Why would running type and screens on the analyzer prevent a patient with a weak D phenotype from forming anti-D?**
- A. Weak D persons cannot form anti-D
- B. The analyzer would show the sample as Rh negative; the patient would receive Rh-negative blood
- C. The analyzer would show the sample as Rh positive; the patient would receive Rh-positive blood
- D. A and C

Blood bank/Correlate clinical and laboratory data/Blood group antigens/2

- 14. A cord blood workup was ordered on baby boy Jones. The mother is O negative. Results on the baby are as follows:**

Anti-A	Anti-B	Anti-A, B	Anti-D	DAT (poly)
4+	0	4+	0	2+

The test for weak D on the baby was positive at the AHG phase. Is the mother an RhIg candidate?

- A. No, the baby is Rh positive
- B. Yes, the baby's Rh type cannot be determined because of the positive DAT result

- C. No, the baby is Rh negative
- D. Yes, the mother is Rh negative

Blood bank/Evaluate laboratory data/Rh type/3

- 15. RBCs from a recently transfused patient were positive on DAT when tested with anti-IgG. Screen cells and a panel performed on a patient's serum showed very weak reactions with inconclusive results. What procedure could help to identify the antibody?**
- A. Elution followed by a panel on the eluate
 - B. Adsorption followed by a panel on the adsorbed serum
 - C. Enzyme panel
 - D. Antigen typing the patient's RBCs

Blood bank/Apply principles of special procedures/Antibody identification/3

Answers to Questions 12–15

- 12. C** Lewis antibodies are usually not clinically significant but may interfere with testing for clinically significant antibodies. Lewis antibodies are most easily removed by neutralizing them with soluble Lewis substance. Lewis antigens are secreted into saliva and plasma and are adsorbed onto RBCs. Le^b substance is made by adding an L-fucose to both the terminal and next to last sugar residue on the type 1 precursor chain. This requires the *Le*, *H*, and *Se* genes. Because some samples of anti-Le^b react only with group O or A₂ RBCs, neutralization is best achieved if the saliva comes from a person who is group O.
- 13. B** The automated analyzer would result the patient with a weak D phenotype as Rh negative, and if blood were needed, the patient would receive Rh-negative blood.
- 14. B** The baby forward types as an A, and the mother is O negative. It is possible that anti-A,B from the mother is attaching to the baby's RBCs, causing a positive DAT result. In the presence of a positive DAT, a weak test for D is not valid. Therefore, the baby's Rh type is unknown, and the mother would be a candidate for RhIg.
- 15. A** If the RBCs show a positive DAT result, then IgG antibody has coated incompatible, antigen-positive RBCs. If screening cells and panel cells show missing or weak reactions, most of the antibody is on the RBCs and would need to be eluted before it can be detected. An elution procedure followed by a panel performed on the eluate would help to identify the antibody.

- 16. A patient types as O positive. All three screening cells and RBCs from two O-positive donor units show agglutination after incubation at 37°C and increase in reactivity at the IAT phase of testing. What action should be taken next?**

- A. Perform an autocontrol and DAT on the patient
- B. Perform an enzyme panel
- C. Perform an elution
- D. Choose another 2 units and repeat crossmatching

Blood bank/Select course of action/Incompatible crossmatch/3

- 17. Four units of blood are ordered for a patient. Blood bank records are checked and indicate that 5 years ago this patient had an anti-Jk^b. What is the next course of action?**

- A. Antigen type units for the Jk^b antigen, and only crossmatch units positive for Jk^b
- B. Antigen type units for the Jk^b antigen, and only crossmatch units negative for Jk^b
- C. Randomly pull 4 units of blood that are ABO compatible, and perform crossmatching
- D. Perform IS crossmatching on 4 Jk^b-negative units

Blood bank/Apply principles of laboratory operations/Compatibility testing/3

- 18. A 56-year-old patient diagnosed with colon cancer demonstrates a positive antibody screen in all three screen cells at the AHG phase. The panel study shows 10 cells as positive as well as the autocontrol at the AHG phase. The reactions varied from 1+ to 3+. This patient had a history of receiving 2 units of blood approximately 1 month ago. What should be done next?**

- A. Perform a DAT on patient cells
- B. Perform an autoadsorption
- C. Perform an alloadsorption
- D. Issue O-negative cells

Blood bank/Evaluate laboratory data to determine best course of action/Panel study/3

- 19. A 33-year-old maternity patient has blood drawn for a type and screen at 36 weeks' pregnancy. The following results are found on the automated blood bank analyzer:**

Anti-A	Anti-B	Anti-A,B	Anti-D	A ₁ cells	B cells
3+	0	4+	4+	2+	4+
SC I	SC II	SC III	A ₁ lectin		
0	0	0	3+		

The reference laboratory identified anti-P₁ in the patient plasma by using enzyme techniques. How could the ABO discrepancy be solved?

- A. Wash the patient's RBCs, and repeat the forward grouping
- B. Test the patient's plasma against A₂ cells
- C. Warm the patient's plasma at 37°C for 10 minutes, and repeat the reverse grouping
- D. Treat the A₁ cells with DTT, and repeat the reverse grouping

Blood bank/Evaluate laboratory data to determine possible inconsistent results/ABO/3

Answers to Questions 16–19

- 16. A** All screening cells and all units are positive at both 37°C and the IAT phase. This indicates the possibility of a high-frequency alloantibody or a warm autoantibody. An autocontrol would help make this distinction. A positive autocontrol indicates an autoantibody is present; a negative autocontrol and positive screen cells indicate an alloantibody. A DAT would be performed to determine if an antibody has coated the patient's RBCs and is directed against screening cells and donor cells.
- 17. B** A patient with a history of a significant antibody, such as anti-Jk^b, must receive blood that has been completely crossmatched and is negative for the corresponding antigen; otherwise, an anamnestic reaction may occur with subsequent lysis of donor cells.
- 18. C** In this situation, an allogeneic adsorption must be performed to adsorb out the autoantibody and leave potential alloantibodies in the patient's serum that will need to

be identified before transfusion of blood to the patient. An autoadsorption cannot be performed because any alloantibodies would be absorbed by circulating donor cells present from the previous month.

19. **C** Anti-P₁ is a cold-reacting antibody. Warming the plasma at 37°C will dissipate the antibody, preventing its reactivity with P₁ antigen on the A₁ cells.

20. **An O-negative mother with no record of any previous pregnancies gives birth to her first child, a B-positive baby. The baby's DAT is weakly positive, and the saline control is negative. The antibody screen is also negative. The baby appears healthy but after 2 days develops mild jaundice, which is treated with phototherapy. After 4 days in the hospital, the baby goes home, without complications. What is the most likely explanation for the weakly positive DAT result?**

- A. Technical error
- B. A low-titer anti-D
- C. Immune anti-B from the mother
- D. A maternal antibody against a low-incidence antigen

Blood bank/Correlate clinical and laboratory data/Hemolytic disease of the fetus and newborn/2

Answer to Question 20

20. **C** In this case, the maternal anti-A,B is probably coating the infant's B cells, causing a positive DAT and jaundice. Anti-A,B from an O person is a single entity that cannot be separated. It is IgG and can cross the placenta. This antibody may attach to A, B, or AB RBCs.

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